FUNCTION SPACES

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Fifth Semester

Contents

1 FUNCTION SPACES 1
- 1.1 An Introduction . . . 1
- 1.2 Metric and Normed Spaces . . . 2
- 1.3 The Lebesgue Integral . . . 3
  - 1.3.1 Upper Functions and L^1 . . . 5
  - 1.3.2 Convergence and Density . . . 8
  - 1.3.3 Integration on Unbounded Intervals . . . 12
  - 1.3.4 Measurable Functions . . . 13

2 THE FOURIER TRANSFORM 15
- 2.1 Convolution . . . 17
- 2.2 Algebras of Functions . . . 21
- 2.3 Fourier Series . . . 23
- 2.4 Hilbert Space and Orthonormality . . . 25
  - 2.4.1 Pointwise Convergence of Fourier Series . . . 28

3 SPACE OF CONTINUOUS FUNCTIONS 31
- 3.1 The Space $C[0,1]$ . . . 31

Index 35

Chapter 1

FUNCTION SPACES

July 20th.

Quizzes 1 and 2 will account for 10% and 15% of the final grade respectively. The mid-term will account for 25% of the final grade, and the final exam will account for 50%.

1.1 An Introduction

Suppose $\mathbb{K}$ is a field, and X is a topological space. Then the set of all $\mathbb{K}$-valued functions on X, denoted by $\mathbb{K}^X$, is a vector space over $\mathbb{K}$. If we let $X = \{1, 2, \ldots, n\}$, then $\mathbb{K}^X \cong \mathbb{K}^n$, the n-dimensional vector space over $\mathbb{K}$. In this course, we will be primarily concerned with the case when $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$. Consider the map

$$D : \mathbb{K}^{\{1,2,\ldots,n\}} \to \mathbb{K}^{\{1,2,\ldots,n-1\}}, \quad (Df)(k) = f(k+1) - f(k) \text{ for } 1 \leq k \leq n-1, \ (Df)(n) = f(1) - f(n). \tag{1.1}$$

Here, D acts as a discrete analogue of the derivative operator by taking finite differences. We can split this finite difference map as the difference of the left shift operator and the identity operator as $D = S - I$. Since S is the left shift operator, we have $S^n = I$ so the eigenvalues of S are a subset of the n-th roots of unity. Let λ be such an eigenvalue. Then a corresponding eigenvector can be given as $\begin{pmatrix} 1 \\ \lambda \\ \vdots \\ \lambda^{n-1} \end{pmatrix}$. If $\lambda = e^{2\pi i/n}$, then we can denote the corresponding eigenvector of λ^k as χ_k. In fact, in the basis $\{\chi_1, \ldots, \chi_{n-1}, \chi_n\}$, D is a diagonal matrix. Such a basis is termed a *discrete Fourier basis*, and the coordinate vector of D with respect to this basis is called the *discrete Fourier transform* of D. The Fourier transform diagonalizes the finite differential operator. We can view χ_k as a function $\chi_k : \{1, 2, \ldots, n\} \to \mathbb{C}$ defined by $\chi_k(j) = e^{2\pi ik(j-1)/n}$ for $1 \leq j \leq n$.

Recall the Hermitian inner product on $\mathbb{C}^n$ given by $\langle x, y \rangle = \sum_{i=1}^n x_i \overline{y_i}$. Viewing $\mathbb{C}^n$ as the function space $\mathbb{C}^{\{1,2,\ldots,n\}}$, we can endow it with the inner product $\langle f, g \rangle = \sum_{k=1}^n f(k)\overline{g(k)}$, which gives the 2-norm $\|f\|_2 = \sqrt{\sum_{i=1}^n |f(i)|^2}$. The discrete Fourier basis is an orthogonal basis with respect to this inner product. Normalizing via division by $\sqrt{n}$ makes it an orthonormal basis. Thus, if we have $f \in \mathbb{C}^{\{1,2,\ldots,n\}}$ which we wish to write as $f = \sum_{j=1}^n a_j \chi_j$, then we can compute the coefficients a_j as

$$a_j = \frac{1}{n}\langle f, \chi_j \rangle = \frac{1}{n} \sum_{r=1}^n f(r) e^{-2\pi ij(r-1)/n}. \tag{1.2}$$

This is the discrete Fourier transform, as before. If we look at the norm, then

$$\|f\|_2^2 = \left\| \sum_{j=1}^n a_j \chi_j \right\|_2^2 = n \sum_{j=1}^n |a_j|^2 \implies \frac{1}{n} \|f\|_2^2 = \sum_{j=1}^n |a_j|^2 . \tag{1.3}$$

This is known as *Plancherel's theorem*.

All of the above can be generalized to the continous case; let $L^2[0,1]$ be the space of square integrable functions on $[0,1]$. If $(a_n)_{n\in\mathbb{Z}}$ is the Fourier series of f, then $\int_0^1 |f(x)|^2\,dx = \sum_{n\in\mathbb{Z}} |a_n|^2$, which is the continuous analogue of Plancherel's theorem.

1.2 Metric and Normed Spaces

July 23rd.

The following theorem relates the notion of compactness, that every cover has a finite subcover, to many other properties of metric spaces.

Theorem 1.1. *Let (X,d) be a metric space. The following are equivalent:*

1. *X is compact.*
2. *Every nested sequence of nonempty closed sets $F_1 \supseteq F_2 \supseteq \cdots$ has nonempty intersection, that is,*
$$\bigcap_{n=1}^{\infty} F_n \neq \emptyset. \tag{1.4}$$
3. *Every countable family of closed sets with the finite intersection property has nonempty intersection.*
4. *Every sequence in X has a convergent subsequence.*
5. *X is complete and totally bounded.*

Proof. For $1 \implies 2$, let $\{F_n\}_{n=1}^{\infty}$ be a nested sequence of nonempty closed sets with $F_1 \supseteq F_2 \supseteq \cdots$. If $\bigcap_{n=1}^{\infty} F_n = \emptyset$, then $\{X \setminus F_n\}_{n=1}^{\infty}$ is an open cover of X. By compactness, there exists a finite subcover. Since the F_n are nested,
$$X = \bigcup_{i=1}^{N} (X \setminus F_i) = X \setminus F_N. \tag{1.5}$$
Hence $F_N = \emptyset$, a contradiction.

For $2 \implies 3$, let $\{F_n\}_{n=1}^{\infty}$ be a countable family of closed sets with the finite intersection property. Define
$$G_n = \bigcap_{i=1}^{n} F_i. \tag{1.6}$$
Then $G_1 \supseteq G_2 \supseteq \cdots$ is a nested sequence of nonempty closed sets. By 2, $\bigcap_{n=1}^{\infty} G_n \neq \emptyset$. Since
$$\bigcap_{n=1}^{\infty} G_n = \bigcap_{n=1}^{\infty} F_n, \tag{1.7}$$
the result follows.

For $3 \implies 4$, let S_0 be the set of elements in $(x_i)_{i=1}^{\infty}$, a sequence in X. Similarly, let S_1 be the set of elements in $(x_i)_{i=2}^{\infty}$, and define S_n similarly. Then
$$\overline{S}_0 \supseteq \overline{S}_1 \supseteq \cdots \supseteq \overline{S}_n \supseteq \cdots. \tag{1.8}$$
By 3, $\bigcap_{n=1}^{\infty} \overline{S}_n \neq \emptyset$, so there exists $x' \in \bigcap_{n=1}^{\infty} \overline{S}_n$. Now, $B(x',1) \cap S_1$ cannot be empty, so pick x_{n_1} from this intersection. Again, $B(x',1/2) \cap S_{n_1}$ cannot be empty, so pick x_{n_2} from this intersection. Continue inductively to obtain the x_{n_k}'s. One can verify that $x_{n_k} \to x'$.

For $4 \implies 5$, every Cauchy sequence has a convergent subsequence, and hence converges. Thus, X is complete. Suppose X is not totally bounded. Then there exists $\alpha > 0$ such that X has no finite covering by balls of radius α. Pick $x_1 \in X$. Since $B(x_1,\alpha) \neq X$, pick $x_2 \in X \setminus B(x_1,\alpha)$. Again,
$$B(x_1,\alpha) \cup B(x_2,\alpha) \neq X. \tag{1.9}$$

Similarly, pick x_3 outside the previous balls, and continue inductively. Since $d(x_m, x_n) \geq \alpha$ for $m \neq n$, there is no convergent subsequence, contradicting 4. Hence X is totally bounded.

For $5 \implies 1$, we prove the contrapositive. Assume X is not compact. If it is not totally bounded, we are done; further assume that it is totally bounded. Then there exists an open cover $\{U_\alpha\}_{\alpha\in\Lambda}$ with no finite subcover. Let $\varepsilon = 1/2$ and pick a finite ε-net. Pick x_1 such that $B(x_1, 1/2)$ is not covered by finitely many U_α. Since $B(x_1, 1/2)$ is totally bounded, pick $x_2 \in B(x_1, 1/2)$ such that $B(x_2, 1/4)$ cannot be covered by finitely many U_α. Inductively, pick $x_n \in B(x_{n-1}, 1/2^{n-1})$ such that $B(x_n, 1/2^n)$ cannot be covered by finitely many U_α. Then

$$d(x_n, x_m) \leq \sum_{k=m}^{n-1} \frac{1}{2^k}, \tag{1.10}$$

so (x_n) is a Cauchy sequence. Since X is complete, $x_n \to x$ for some $x \in X$. Let $x \in U_\alpha$. Then there exists $\varepsilon > 0$ such that $B(x, \varepsilon) \subseteq U_\alpha$. Moreover, $d(x_n, x) < \varepsilon/2$ for n sufficiently large. Hence

$$B(x_n, 2^{-n}) \subseteq B(x_n, \varepsilon/2) \subseteq B(x, \varepsilon) \subseteq U_\alpha, \tag{1.11}$$

for all sufficiently large n, since $2^{-n} < \varepsilon/2$. This contradicts the choice of $B(x_n, 2^{-n})$. Hence X is not complete. ■

We now look at some examples of non-trivial function spaces; consider $C[0,1]$, the space of complex-valued continuous functions on $[0,1]$. This is a vector space over $\mathbb{R}$ with pointwise addition and scalar multiplication. We can define a norm on this space as $\|f\|_\infty = \sup_{x\in[0,1]} |f(x)|$. This makes $C[0,1]$ into a normed space.

Definition 1.2. Let X be a $\mathbb{K}$-vector space. A *norm* on X is a function $\|\cdot\| : X \to [0,\infty)$ such that for all $x, y \in X$ and $\alpha \in \mathbb{K}$,

1. $\|x\| = 0$ if and only if $x = 0$ (positive definiteness).
2. $\|\alpha x\| = |\alpha| \, \|x\|$ (homogeneity).
3. $\|x + y\| \leq \|x\| + \|y\|$ (triangle inequality).

One may verify that $\|\cdot\|_\infty$ is indeed a norm on $C[0,1]$.

1.3 The Lebesgue Integral

July 27th.

Definition 1.3. A function $s : [0,1] \to \mathbb{C}$ is called a *step function* if there exists a partition $P : 0 = x_0 < x_1 < \cdots < x_n = 1$ of $[0,1]$ such that s is constant on each interval (x_{i-1}, x_i) for $1 \leq i \leq n$. That is, $s(x) = c_k$ if $x \in (x_{k-1}, x_k)$ where c_k are constants for $1 \leq k \leq n$.

We impose that the integral of such a step function is given by

$$\int_0^1 s(x)\,dx = \sum_{k=1}^{n} c_k(x_k - x_{k-1}). \tag{1.12}$$

The values of the step function on the partition points x_i do not affect the value of the integral.

Proposition 1.4. *The set of step functions on $[0,1]$ is a complex vector space under pointwise addition and scalar multiplication.*

The proof is straightforward and left as an exercise.

Proposition 1.5. *The set of step functions on* $[0,1]$ *such that* $\int_0^1 |s(x)|\,dx = 0$ *is a subspace of the vector space of step functions on* $[0,1]$.

Again, this proof is rudimentary and left as an exercise. The reader is also urged to show that $\int_0^1 s(x)\,dx$ is well-defined and independent of the choice of partition P.

Definition 1.6. A set $S \subseteq [0,1]$ is said to have *content zero* or measure zero if for every $\varepsilon > 0$, there exists countably many open intervals $(a_n, b_n)_{n=1}^{\infty}$ such that $S \subseteq \bigcup_{n=1}^{\infty}(a_n, b_n)$ and $\sum_{n=1}^{\infty}(b_n - a_n) < \varepsilon$.

As a trivial example, any finite subset of $[0,1]$ has content zero. Similarly, any countable subset of $[0,1]$ has content zero. The Cantor set is an example of an uncountable set with content zero. One may show that the countable union of sets of content zero has content zero.

Definition 1.7. A sequence of functions $f_n : [0,1] \to \mathbb{R}$ is said to be increasing if $f_n(x) \leq f_{n+1}(x)$ for all $x \in [0,1]$. Similarly, a sequence of decreasing functions is defined.

Note that a property is said to be hold *almost everywhere* if it holds for all $x \in [0,1]$ except for a negligible set (a set of content zero).

Definition 1.8. We say $f_n \to f$ pointwise almost everywhere if there exists a set $S \subseteq [0,1]$ of content zero such that $f_n(x) \to f(x)$ for all $x \in [0,1] \setminus S$. That is, f_n converges to f pointwise except on a negligible set.

Theorem 1.9 (The *proto monotone convergence theorem*)**.** *Let* $\{s_n\}_{n=1}^{\infty}$ *be a decreasing seqeucence of non-negative step functions on* $[0,1]$ *such that* $s_n \to 0$ *pointwise almost everywhere. Then* $\int_0^1 s_n(x)\,dx \to 0$ *as* $n \to \infty$.

Proof. Let $E := \{x \in [0,1] \mid s_n(x) \not\to 0\}$. Then E has content zero. Define D to be the set of endpoints of intervals in the partitions corresponding to the s_n's. Then D is countable, and hence has content zero. Let $F = D \cup E$ is negligible. Let $x \in [0,1] \setminus F$. Then there exists $N(x) \in \mathbb{N}$ such that $s_n(x) = 0$ for all $n \geq N(x)$. Since $x \notin D$, there exists an open interval $B(x)$ such that $s_N(t) < \varepsilon$ for all $t \in B(x)$. As $\{s_n\}$ is decreasing, we have $s_n(t) < \varepsilon$ for all $t \in B(x)$ and $n \geq N(x)$. For every $x \in [0,1] \setminus F$, we have a $B(x)$. Find a sequence of open intervals such that $F \subseteq \bigcup_{n=1}^{\infty} F_n$ and $\sum_{n=1}^{\infty} |F_n| < \varepsilon$. Then

$$[0,1] \subseteq \left(\bigcup_{x \in [0,1]\setminus F} B(x) \right) \cup \left(\bigcup_{n=1}^{\infty} F_n \right), \tag{1.13}$$

an open cover. By compactness, there exists a finite subcover, so

$$[0,1] \subseteq \bigcup_{i=1}^{p} B(x_i) \cup \bigcup_{r=1}^{q} F_r. \tag{1.14}$$

Call the left union P and the right union Q. Then

$$\int_Q s_n(x)\,dx := \int_0^1 \chi_Q(x) s_n(x)\,dx \leq M \int \sum_{i=1}^{q} |F_i|\,dx < M\varepsilon. \tag{1.15}$$

Similarly,

$$\int_P s_n(x)\,dx := \int_0^1 \chi_P(x) s_n(x)\,dx \leq \varepsilon \quad \text{for all } n \geq \max\{N(x_i) \mid 1 \leq i \leq p\}. \tag{1.16}$$

Combining gives

$$\int_0^1 s_n(x)\,dx = \int_P s_n(x)\,dx + \int_Q s_n(x)\,dx < (M+1)\varepsilon. \tag{1.17}$$

Thus, $\int_0^1 s_n(x)\,dx \to 0$ as $n \to \infty$. ∎

From here, we can immediately note that $\rho : \mathcal{S}[0,1] \to \mathbb{C}$, map from the set of step functions, defined by $s \mapsto \int_0^1 s(x)\,dx$ is a linear functional. We can also define a semi-norm on the set of step functions as $\|s\|_1 = \int_0^1 |s(x)|\,dx$. This is a semi-norm since $\|s\|_1 = 0$ does not imply that $s = 0$; it only implies that $s = 0$ almost everywhere. We can define an equivalence relation on the set of step functions by $s \sim t$ if and only if $s = t$ almost everywhere. Then we can define a norm on the quotient space $\mathcal{S}[0,1]/\sim$ as $\|s\|_1 = \int_0^1 |s(x)|\,dx$. This is indeed a norm since $\|s\|_1 = 0$ implies that $s = 0$ in the quotient space.

The normed space $(\mathcal{S}[0,1]/\sim, \|\cdot\|_1)$ is not complete; there is a much larger class of functions that are (Lebesgue) integrable. Henceforth, we define $L^1[0,1]$ to be the completion of $(\mathcal{S}[0,1]/\sim, \|\cdot\|_1)$. The elements of $L^1[0,1]$ are called *Lebesgue integrable functions*.

July 30th.

Theorem 1.10. *Let $\{t_n\}_{n=1}^\infty$ be a sequence of step functions on $[0,1]$ such that*

1. *There is a function f such that $t_n \uparrow f$ pointwise almost everywhere.*
2. *The sequence $\{\int_0^1 t_n(x)\,dx\}_{n=1}^\infty$ converges.*

Then, for any step function t such that $t(x) \le f(x)$ almost everywhere, we have $\int_0^1 t(x)\,dx \le \lim_{n\to\infty} \int_0^1 t_n(x)\,dx$.

Proof. Define s_n as

$$s_n(x) = \begin{cases} t(x) - t_n(x), & \text{if } t(x) \ge t_n(x), \\ 0, & \text{if otherwise.} \end{cases} \tag{1.18}$$

That is, $s_n(x) = \max\{t(x) - t_n(x), 0\}$ for all $n \ge 1$. Also, $s_n(x) \ge t(x) - t_n(x)$ almost everywhere, so

$$\int_0^1 s_n(x)\,dx \ge \int_0^1 t(x)\,dx - \int_0^1 t_n(x)\,dx. \tag{1.19}$$

Taking limits on both sides gives

$$\lim_{n\to\infty} \int_0^1 t_n(x)\,dx \ge \int_0^1 t(x)\,dx \tag{1.20}$$

since $\lim_{n\to\infty} \int_0^1 s_n(x)\,dx = 0$ by the proto monotone convergence theorem. ∎

1.3.1 Upper Functions and L^1

Recall the linear functional $\rho : \mathcal{S}[0,1] \to \mathbb{R}$ (or $\mathbb{C}$) defined by $s \mapsto \int_0^1 s(x)\,dx$. We can extend this functional to a larger class of functions.

Definition 1.11. A function $f : [0,1] \to \mathbb{R}$ is said to be an upper function on $[0,1]$ if there exists an increasing sequence of step functions $\{s_n\}_{n=1}^\infty$ such that $s_n \uparrow f$ pointwise almost everywhere, and $\int_0^1 s_n(x)\,dx$ converges.

At this point, we can define the integral of an upper function f as $\int_0^1 f dx := \lim_{n\to\infty} \int_0^1 s_n(x)\,dx$. The right hand side is well-defined by the previous theorem, but we need to show that it is independent of the choice of the sequence $\{s_n\}_{n=1}^\infty$ that converges to f pointwise almost everywhere. This is the content of the following theorem.

Theorem 1.12. *$\int_0^1 f dx$ is well-defined for an upper function f.*

Proof. Let $\{s_n\}_{n=1}^{\infty}$ and $\{t_n\}_{n=1}^{\infty}$ be two increasing sequences of step functions such that $s_n \uparrow f$ and $t_n \uparrow f$ pointwise almost everywhere. Since $s_n \leq f$ almost everywhere, we have $\int_0^1 s_n(x)\,dx \leq \lim_{k\to\infty} \int_0^1 t_k(x)\,dx$ by the previous theorem. Thus,

$$\lim_{n\to\infty} \int_0^1 s_n(x)\,dx \leq \lim_{k\to\infty} \int_0^1 t_k(x)\,dx. \tag{1.21}$$

By a symmetry argument, we have $\lim_{n\to\infty} \int_0^1 t_n(x)\,dx = \lim_{k\to\infty} \int_0^1 s_k(x)\,dx$. Hence, $\int_0^1 f dx$ is well-defined. ■

Let $U[0,1]$ denote the set of upper functions on $[0,1]$. Trivially, $\mathcal{S}[0,1] \subseteq U[0,1]$.

Theorem 1.13. *Let $f, g \in U[0,1]$, and $\lambda \geq 0$. Then $f+g \in U[0,1]$ and $\lambda f \in U[0,1]$. Moreover,*

$$\int_0^1 (f+g)\,dx = \int_0^1 f\,dx + \int_0^1 g dx, \quad \int_0^1 (\lambda f)\,dx = \lambda \int_0^1 f\,dx. \tag{1.22}$$

If $f \leq g$ almost everywhere, then $\int_0^1 f dx \leq \int_0^1 g dx$.

Note that it is clear that all these properties are true for step functions.

Proof. Suppose $s_n \uparrow f$ and $t_n \uparrow g$ almost everywhere, and that $\int_0^1 s_n(x)\,dx$ and $\int_0^1 t_n(x)\,dx$ converge. Then $s_n + t_n \uparrow f+g$ almost everywhere, and $\lambda s_n \uparrow \lambda f$ almost everywhere. By the linearity of the integral for step functions, we have

$$\int_0^1 (s_n + t_n)\,dx = \int_0^1 s_n\,dx + \int_0^1 t_n\,dx, \quad \int_0^1 (\lambda s_n)\,dx = \lambda \int_0^1 s_n\,dx. \tag{1.23}$$

Now suppose $f \leq g$ almost everywhere. Then $s_n \leq g$ almost everywhere, so $\int_0^1 s_n(x)\,dx \leq \int_0^1 g\,dx$ by the previous theorem. Taking limits gives $\int_0^1 f\,dx \leq \int_0^1 g\,dx$. ■

Theorem 1.14. *If $f, g \in U[0,1]$ and $f = g$ almost everywhere, then $\int_0^1 f\,dx = \int_0^1 g\,dx$.*

Proof. Note that $f \leq g$ almost everywhere and $g \leq f$ almost everywhere. Hence, $\int_0^1 f\,dx \leq \int_0^1 g\,dx$ and $\int_0^1 g\,dx \leq \int_0^1 f\,dx$. Thus, $\int_0^1 f\,dx = \int_0^1 g\,dx$. ■

Theorem 1.15. *If $f, g \in U[0,1]$, then $\max\{f,g\}, \min\{f,g\} \in U[0,1]$.*

August 3rd.

Theorem 1.16. *Let $f : [0,1] \to \mathbb{R}$ be continuous and bounded. Then $f \in U[0,1]$ and the Lebesgue integral of f is equal to the Riemann integral of f.*

Proof. Let $\mathcal{P}_n : x_0 < \cdots < x_{2^n}$ be a partition of $[0,1]$ into 2^n equal subintervals of length 2^{-n}. Let $m_k = \inf\{f(x) \mid x \in [x_{k-1}, x_k]\}$ for $1 \leq k \leq 2^n$, and define the step function s_n as

$$s_n(x) = \begin{cases} m_k, & \text{if } x \in (x_{k-1}, x_k], \\ m_1, & \text{if } x = x_0. \end{cases} \tag{1.24}$$

Infer that $s_{n+1} \geq s_n$ since

$$\inf\{f(x) \mid x \in [x_{k-1}, x_k]\} \leq \inf\{f(x) \mid x \in [x_{2k-2}, x_{2k}]\} \leq \inf\{f(x) \mid x \in [x_{2k-2}, x_{2k-1}]\}. \tag{1.25}$$

Also observe that $s_n(x) \to f(x)$ for all $x \in [0,1]$ for which f is continuous. Given $\varepsilon > 0$, there exists $\delta > 0$ such that for all $y \in (x-\delta, x+\delta) \cap [0,1]$, we have $|f(y) - f(x)| < \varepsilon$. Choose N such that the partition $\mathcal{P}_N$ has a subinterval $[x_{k-1}, x_k]$ containing x and lying in $(x-\delta, x+\delta)$. Then

$$s_N(x) = m_k \leq f(x) \leq \inf\{f(y) \mid y \in (x-\delta, x+\delta)\} \leq m_k + \varepsilon = s_N(x) + \varepsilon. \tag{1.26}$$

That is, $|s_N(x) - f(x)| < \varepsilon$. Since ε was arbitrary, $s_n(x) \to f(x)$ for all $x \in [0,1]$. Finally, observe that $\int_0^1 s_n(x)\,dx$ is the lower Riemann sum of f with respect to the partition $\mathcal{P}_n$. Since f is Riemann integrable, we have $\int_0^1 s_n(x)\,dx \to \int_0^1 f(x)\,dx$ as $n \to \infty$. Hence, $f \in U[0,1]$ and the Lebesgue integral of f is equal to the Riemann integral of f. ■

Given general upper functions $u, v \in U[0,1]$, the function $u - v$ may not neccesarily lie in $U[0,1]$. From here, we define the following.

Definition 1.17. $L^1[0,1]$ is the set of functions $f : [0,1] \to \mathbb{R}$ such that $f = u - v$ for some $u, v \in U[0,1]$. It is termed the space of Lebesgue integrable functions on $[0,1]$. f is said to be *Lebesgue integrable* here, with the Lebesgue integral of f defined as

$$\int_0^1 f\,dx := \int_0^1 u\,dx - \int_0^1 v\,dx. \tag{1.27}$$

Theorem 1.18. *Let $u, v, w, z \in U[0,1]$ be such that $u - v = w - z$ almost everywhere. Then $\int_0^1 u\,dx - \int_0^1 v\,dx = \int_0^1 w\,dx - \int_0^1 z\,dx$.*

Proof. Note that $u + z = w + v$ almost everywhere. Hence, $\int_0^1 u\,dx + \int_0^1 z\,dx = \int_0^1 w\,dx + \int_0^1 v\,dx$. Rearranging gives the desired result. ■

Thus, the Lebesgue integral of $f \in L^1[0,1]$ is well-defined.

Theorem 1.19. *Let $f, g \in L^1[0,1]$. Then*

1. *$af + bg \in L^1[0,1]$ for all $a, b \in \mathbb{R}$, and $\int_0^1 (af + bg)\,dx = a\int_0^1 f\,dx + b\int_0^1 g\,dx$.*
2. *$\int_0^1 f\,dx \geq 0$ if $f \geq 0$ almost everywhere.*
3. *$\int_0^1 f\,dx \geq \int_0^1 g\,dx$ if $f \geq g$ almost everywhere.*
4. *$\int_0^1 f\,dx = \int_0^1 g\,dx$ if $f = g$ almost everywhere.*

Proof.

1. Let $a^+ = \max\{a, 0\}$ and $a^- = \max\{-a, 0\}$. Then $af = a(u - v) = au - av = (a^+u + a^-v) - (a^-u + a^+v)$. Similarly, $bg = b(w - z) = (b^+w + b^-z) - (b^-w + b^+z)$. Hence, $af + bg \in L^1[0,1]$. The linearity of the integral follows from the linearity of the integral on upper functions.

2. Let $f = u - v$ for some $u, v \in U[0,1]$. Then $u \geq v$ almost everywhere, so $\int_0^1 u\,dx \geq \int_0^1 v\,dx$. Hence, $\int_0^1 f\,dx = \int_0^1 u\,dx - \int_0^1 v\,dx \geq 0$.

3. Follows from 2.

4. Follows from 2 and 3.

■

If f is a real-valued function, we can define the positive and negative parts of f as $f^+ = \max\{f, 0\}$ and $f^- = \max\{-f, 0\}$. Then $f = f^+ - f^-$ and $|f| = f^+ + f^-$.

Theorem 1.20. *If $f, g \in L^1[0,1]$, then $f^+, f^-, |f|, \max\{f, g\}, \min\{f, g\} \in L^1[0,1]$. Moreover,*

$$\left|\int_0^1 f\,dx\right| \leq \int_0^1 |f|\,dx. \tag{1.28}$$

Proof. Let $f = u - v$ for some $u, v \in U[0,1]$. Then $f^+ = \max\{u - v, 0\} = \max\{u, v\} - v \in L^1[0,1]$ and $f^- = \max\{-f, 0\} = \max\{v - u, 0\} = \max\{u, v\} - u \in L^1[0,1]$. Hence, $|f| = f^+ + f^- \in L^1[0,1]$. Similarly, $\max\{f, g\}, \min\{f, g\} \in L^1[0,1]$. Finally,

$$-\int_0^1 |f|\,dx \le -\int_0^1 f^-\,dx = \int_0^1 f^+ - f^-\,dx = \int_0^1 f\,dx \le \int_0^1 f^+\,dx \le \int_0^1 |f|\,dx. \tag{1.29}$$

■

August 6th.

We also recall the following theorem from real analysis.

Theorem 1.21 (*Lebesgue-Vitali theorem*)**.** *$f : [0,1] \to \mathbb{R}$ is Riemann integrable if and only if f is continuous almost everywhere, and bounded.*

Theorem 1.16 is a generalization of this, with the condition of f being continuous replaced with f being continuous almost everywhere. Of course, this theory extends to complex-valued functions.

Definition 1.22. A function $f : [0,1] \to \mathbb{C}$ is said to be Lebesgue integrable, denoted $f \in L^1([0,1];\mathbb{C})(\equiv L^1[0,1])$ if $\Re(f), \Im(f) \in L^1([0,1];\mathbb{R})$. The Lebesgue integral of f is defined as

$$\int_0^1 f\,dx := \int_0^1 \Re(f)\,dx + i\int_0^1 \Im(f)\,dx \tag{1.30}$$

where $\Re(f) = \frac{f+\overline{f}}{2}$ and $\Im(f) = \frac{f-\overline{f}}{2i}$.

Theorem 1.23. *If $f \in L^1([0,1];\mathbb{C})$, then $|f| \in L^1([0,1];\mathbb{R})$ and $\left|\int_0^1 f\,dx\right| \le \int_0^1 |f|\,dx$.*

Proof. Note that $h_1 = \Re(f), h_2 = \Im(f) \in L^1([0,1];\mathbb{R})$, so $|\Re(f)|, |\Im(f)| \in L^1([0,1];\mathbb{R})$. Let s_n and t_n be a sequence of step functions that converge to h_1 and h_2 from below. Then,

$$\sqrt{s_n^2 + t_n^2} \le |s_n| + |t_n| \implies \int_0^1 |f(x)|\,dx := \lim_{n\to\infty}\int_0^1 \sqrt{s_n^2 + t_n^2}\,dx \le \int_0^1 |h_1(x)|\,dx + \int_0^1 |h_2(x)|\,dx < \infty. \tag{1.31}$$

Thus, $|f| \in L^1([0,1];\mathbb{R})$. For the inequality, we can break down f as $f(x) = u(x)\,|f(x)|$, where $|u(x)| = 1$ for all $x \in [0,1]$. Thus, $\overline{u(x)}f(x) = |f(x)|$ ■

1.3.2 Convergence and Density

Recall the equivalence relation $\sim$ used to define the quotient space $\mathcal{S}[0,1]/\sim$, which was then extended to form $L^1[0,1]$. We claim that $\mathcal{S}[0,1]/\sim \subseteq L^1[0,1]$ is dense in $(L^1[0,1], \|\cdot\|_1)$.

Lemma 1.24. *$(\mathcal{S}[0,1]/\sim, \|\cdot\|_1)$ is dense in $(C[0,1], \|\cdot\|_1)$.*

Proof. Let $f \in C[0,1]$. Then f is bounded, so there exists $M > 0$ such that $|f(x)| \le M$ for all $x \in [0,1]$. Let $\varepsilon > 0$. By the uniform continuity of f, there exists $\delta > 0$ such that $|f(x) - f(y)| < \varepsilon$ whenever $|x - y| < \delta$. Choose $N \in \mathbb{N}$ such that $2^{-N} < \delta$. Consider the partition $\mathcal{P}_N : x_0 < x_1 < \cdots < x_{2^N}$ of $[0,1]$ into 2^N equal subintervals of length 2^{-N}. Define the step function s_N as

$$s_N(x) = f(x_{k-1}) \quad \text{for } x \in (x_{k-1}, x_k], \quad 1 \le k \le 2^N. \tag{1.32}$$

Then,

$$\|f - s_N\|_1 = \int_0^1 |f(x) - s_N(x)|\,dx = \sum_{k=1}^{2^N}\int_{x_{k-1}}^{x_k} |f(x) - f(x_{k-1})|\,dx < 2^N 2^{-N}\varepsilon = \varepsilon. \tag{1.33}$$

Hence, $(\mathcal{S}[0,1]/\sim, \|\cdot\|_1)$ is dense in $(C[0,1], \|\cdot\|_1)$. ■

Lemma 1.25. *$(C[0,1], \|\cdot\|_1)$ is dense in $(L^1[0,1], \|\cdot\|_1)$.*

Theorem 1.26. *Let $\{s_n\}_{n\geq 1}$ be an increasing sequence of step functions such that $\lim_{n\to\infty} \int_0^1 s_n(x)\,dx$ exists. Then s_n converges pointwise almost everywhere to some upper function $f \in U[0,1]$ and $\int_0^1 f\,dx = \lim_{n\to\infty} \int_0^1 s_n(x)\,dx = \int_0^1 \lim_{n\to\infty} s_n(x)\,dx$.*

Proof. Without the loss of generality, we can assume that $s_n \geq 0$ for all $n \geq 1$; also Let $M = \lim_{n\to\infty} \int_0^1 s_n(x)\,dx$. Define $D = \{x \in [0,1] \mid s_n(x) \uparrow \infty \text{ or diverges}\}$. We wish to show that D has content zero. Let $\varepsilon > 0$. Further define $t_n(x) = \left\lfloor \frac{\varepsilon}{2M} s_n(x) \right\rfloor$ for all $x \in [0,1]$ and $n \geq 1$. These t_n are an increasing sequence of step functions. If $\{s_n\}_{n\geq 1}$ converges, then $\{t_n\}_{n\geq 1}$ converges as well and, hence, is eventually constant. If $\{s_n\}_{n\geq 1}$ diverges, then $t_{n+1}(x) - t_n(x) \geq 1$ for infinitely many values of n. Let

$$D_n = \{x \in [0,1] \mid t_{n+1}(x) - t_n(x) \geq 1\}. \tag{1.34}$$

Notice that $D \subseteq \bigcup_{n=1}^{\infty} D_n$. Now,

$$\int_0^1 (t_{n+1}(x) - t_n(x))\,dx \geq \int_0^1 \chi_{D_n}(x)(t_{n+1}(x) - t_n(x))\,dx \geq |D_n|\,. \tag{1.35}$$

This leads to

$$\begin{aligned}\sum_{n=1}^{m} |D_n| &\leq \sum_{n=1}^{m} \int_0^1 (t_{n+1}(x) - t_n(x))\,dx = \int_0^1 t_{m+1}(x)\,dx - \int_0^1 t_1(x)\,dx \\ &\leq \int_0^1 t_{m+1}(x)\,dx \leq \frac{\varepsilon}{2M} \int_0^1 s_{m+1}(x)\,dx < \varepsilon.\end{aligned} \tag{1.36}$$

Thus, $\sum_{n=1}^{\infty} |D_n| < \varepsilon$, and hence $|D| = 0$. The rest of the proof follows from the definition of upper functions. ■

August 10th.

The above is known as the monotone converge theorem for step functions. We can extend this to upper functions as follows.

Theorem 1.27. *Let $\{f_n\}_{n\geq 1}$ be an increasing sequence of upper functions such that $\lim_{n\to\infty} \int_0^1 f_n(x)\,dx$ exists. Then f_n converges pointwise almost everywhere to some upper function $f \in U[0,1]$ and $\int_0^1 f\,dx = \lim_{n\to\infty} \int_0^1 f_n(x)\,dx = \int_0^1 \lim_{n\to\infty} f_n(x)\,dx$.*

Proof. Let $s_{n,k} \uparrow f_k$ almost everywhere, and define $t_n(x) = \max\{s_{n,1}(x), \ldots, s_{n,n}(x)\}$. Then t_n is increasing, and $t_n(x) \leq \max\{f_1(x), \ldots, f_n(x)\} = f_n(x)$ almost everywhere. Therefore, $\lim_0^1 t_n(x)\,dx \leq \int_0^1 f_n(x)\,dx \leq M$ for some M. Define $f(x) := \lim_{n\to\infty} t_n(x)$, wherever the limit exists. By the monotone convergence theorem for step function, f is defined almost everywhere, and $\int_0^1 f\,dx = \lim_{n\to\infty} \int_0^1 t_n(x)\,dx$. Also, $s_{n,k}(x) \leq t_n(x)$ for all $k \leq n$, so $f_k(x) = \lim_{n\to\infty} s_{n,k}(x) \leq \lim_{n\to\infty} t_n(x) = f(x)$ almost everywhere. Let $f_k(x) \to g(x)$ almost everywhere; we show that $f(x) = g(x)$ almost everywhere. Note that $g(x) \leq f(x)$ almost everywhere. Then $t_n(x) \leq f_n(x) \leq g(x)$ almost everywhere, so $f(x) \leq g(x)$ almost everywhere. Hence, $f(x) = g(x)$ almost everywhere. Hence, $f \in U[0,1]$. The rest of the proof follows from the monotone convergence theorem for step functions. ■

Again, we can extend this to Lebesgue integrable functions as follows.

Theorem 1.28 (The *monotone convergence theorem*)**.** *Let $\{f_n\}_{n\geq 1}$ be an increasing sequence of Lebesgue integrable functions such that $\lim_{n\to\infty} \int_0^1 f_n(x)\,dx$ exists. Then f_n converges pointwise almost everywhere to some Lebesgue integrable function $f \in L^1[0,1]$ and $\int_0^1 f\,dx = \lim_{n\to\infty} \int_0^1 f_n(x)\,dx = \int_0^1 \lim_{n\to\infty} f_n(x)\,dx$.*
Alternatively, let $\{g_n\}_{n\geq 1}$ be a sequence of non-negative, almost everywhere, Lebesgue integrable func-

tions such that $\sum_{n=1}^{\infty}\int_0^1 g_n(x)\,dx$ converges. Then the series $\sum_{n=1}^{\infty} g_n(x)$ converges almost everywhere to a Lebesgue integrable function $g \in L^1[0,1]$, and $\int_0^1 g\,dx = \int_0^1 \sum_{n=1}^{\infty} g_n(x)\,dx = \sum_{n=1}^{\infty}\int_0^1 g_n(x)\,dx$.

We require the following theorem to prove the monotone convergence theorem for Lebesgue integrable functions.

Theorem 1.29. *Let $f \in L^1[0,1]$ and $\varepsilon > 0$. Then*

1. *there exist upper functions $u, v \in U[0,1]$ such that $f = u - v$, with v non-negative almost everywhere, and $\int_0^1 v\,dx < \varepsilon$, and*
2. *there exists a step function $s \in \mathcal{S}[0,1]$ and $g \in L^1[0,1]$ such that $f = s + g$ with $\|g\|_1 < \varepsilon$.*

Proof. 1. Let $f = u_1 - v_1$, with $u_1, v_2 \in U[0,1]$. Let $\{t_n\}_{n\geq 1} \subseteq \mathcal{S}[0,1]$ be such that $t_n \uparrow v_1$, and let $N \in \mathbb{N}$ be such that $0 \leq \int_0^1 (v_1 - t_N)(x)\,dx < \varepsilon$. Then $f = u_1 - t_N - (v_1 - t_N)$, with $u = u_1 - t_N$ and $v = v_1 - t_N$.

2. Let $f = u - v$, with $u, v \in U[0,1]$, and $v \geq 0$ almost everywere. Pick a step function $s \in \mathcal{S}[0,1]$ such that $0 \leq \int_0^1 (u-s)(x)\,dx < \frac{\varepsilon}{2}$. Then $f = u - v = s + (u-s) - v$, wth $g = u - s - v \in L^1[0,1]$. Then $\|g\|_1 = \int_0^1 |g(x)|\,dx \leq \int_0^1 (u-s)(x)\,dx + \int_0^1 v(x)\,dx < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$.

■

Proof of monotone convergence theorem. For $\varepsilon > 0$, let $g = u_\varepsilon - v_\varepsilon$, with $u_\varepsilon, v_\varepsilon \in U[0,1]$, $v_\varepsilon \geq 0$ almost everywhere, and $\int_0^1 v_\varepsilon\,dx < \varepsilon$, by the above theorem; for $n \in \mathbb{N}$, there exists a decomposition $g_n = u_n - v_n$, with $u_n, v_n \in U[0,1]$, $v_n \geq 0$ almost everywhere, and $\int_0^1 v_n\,dx < \frac{1}{2^n}$. Then $\sum_{n=1}^{\infty}\left(\int_0^1 v_n(x)\,dx\right)$ converges. Now, $u_n = g + v_n \geq 0$ almost everywhere; define $U_n(x) = \sum_{k=1}^{n} u_k(x)$, an increasing sequence of upper functions.Then

$$\int_0^1 U_n(x)\,dx = \sum_{k=1}^{n}\int_0^1 u_k(x)\,dx = \sum_{k=1}^{n}\left(\int_0^1 g(x)\,dx\right) + \sum_{k=1}^{n}\left(\int_0^1 v_k(x)\,dx\right). \tag{1.37}$$

However, both the sums on the right are finite; one by hypothesis, and one by the choice of the v_k's. Hence, $\lim_{n\to\infty}\int_0^1 U_n(x)\,dx$ exists. By the monotone convergence theorem for upper functions, U_n converges pointwise almost everywhere to U, an upper function on $[0,1]$, and $\int_0^1 U(x)\,dx = \lim_{n\to\infty}\int_0^1 U_n(x)\,dx$. Similarly, one can obtain V with $\int_0^1 V(x)\,dx = \lim_{n\to\infty}\int_0^1 V_n(x)\,dx$. Then, define $g = U - V$, which is Lebesgue integrable, and $\int_0^1 g\,dx = \sum_{n=1}^{\infty}\int_0^1 g_n(x)\,dx$. ■

This naturally leads to the next theorem.
August 17th.

Theorem 1.30 (The *dominated convergence theorem*)**.** *Let $\{f_n\}_{n\geq 1}$ be a sequence of Lebesgue integrable functions such that $f_n \to f$ almost everywhere, and there exists a Lebesgue integrable function $g \geq 0$ almost everywhere such that $|f_n| \leq g$ almost everywhere for all $n \geq 1$. Then $f \in L^1[0,1]$ and $\int_0^1 f\,dx = \lim_{n\to\infty}\int_0^1 f_n(x)\,dx$.*

Proof. Since f converges almost everywhere,

$$\lim_{n\to\infty} f_n(x) = \limsup_{n\to\infty} f_n(x) = \inf_{n\geq 1}\sup_{k\geq n} f_k(x) = \lim_{n\to\infty}\sup_{k\geq n} f_k(x). \tag{1.38}$$

Now define $G_{n,i}(x) = \max\{f_i(x), \ldots, f_{i+n}(x)\}$ for $i, n \geq 1$. Then $|G_{n,i}| \leq g$ almost everywhere, and $\left|\int_0^1 G_{n,i}(x)\,dx\right| \leq \int_0^1 g(x)\,dx < \infty$. So for each fixed i, $\{G_{n,i}\}_{n\geq 1}$ is an increasing sequence of Lebesgue integrable functions. By the monotone convergence theorem, there exists $G_i \in L^1[0,1]$ such that $G_{n,i} \uparrow G_i$ and $\int_0^1 G_i(x)\,dx = \lim_{n\to\infty}\int_0^1 G_{n,i}(x)\,dx$. But note that $G_i(x)$ is simply $\sup_{k\geq i} f_k(x)$. Since $f_n \leq |f_n| \leq g$ almost everywhere, we have $G_i(x) \leq g(x)$ almost everywhere. Since the G_i's are decreasing, the sequence $\{g - G_i\}_{i\geq 1}$ is an increasing sequence of non-negative Lebesgue integrable

functions, with $g - G_i \leq 2g$; by the monotone convergence theorem, there exists $h \in L^1[0,1]$ such that $g - G_i \uparrow h$ and $\int_0^1 h(x)\,dx = \lim_{i\to\infty} \int_0^1 (g - G_i)(x)\,dx$. But note that $h(x) = g(x) - \lim_{i\to\infty} G_i(x) = g(x) - \limsup_{n\to\infty} f_n(x) = g(x) - f(x)$ almost everywhere. Hence, $f = g - h$ is Lebesgue integrable, and

$$\int_0^1 f(x)\,dx = \int_0^1 g(x)\,dx - \int_0^1 h(x)\,dx = \int_0^1 g(x)\,dx - \lim_{i\to\infty} \int_0^1 (g - G_i)(x)\,dx = \lim_{i\to\infty} \int_0^1 G_i(x)\,dx. \tag{1.39}$$

■

Lemma 1.31 (*Fatou's lemma*)**.** *Let $\{f_n\}_{n\geq 1}$ be a sequence of non-negative Lebesgue integrable functions, and let $f(x) = \liminf_{n\to\infty} f_n(x)$. If $\liminf_{n\to\infty} \int_0^1 f_n(x)\,dx < \infty$, then f is Lebesgue integrable and $\int_0^1 f(x)\,dx \leq \liminf_{n\to\infty} \int_0^1 f_n(x)\,dx$.*

Proof. Let $g_n(x) = \inf_{k\geq n} f_k(x)$. Then $\{g_n\}_{n\geq 1}$ is an increasing sequence of non-negative Lebesgue integrable functions, with $g_n(x) \uparrow f(x)$ almost everywhere. By the monotone convergence theorem, there exists $g \in L^1[0,1]$ such that $g_n \uparrow g$ and $\int_0^1 g(x)\,dx = \lim_{n\to\infty} \int_0^1 g_n(x)\,dx$. But note that $g(x) = f(x)$ almost everywhere. Hence, f is Lebesgue integrable, and

$$\int_0^1 f(x)\,dx = \lim_{n\to\infty} \int_0^1 g_n(x)\,dx = \liminf_{n\to\infty} \int_0^1 f_n(x)\,dx. \tag{1.40}$$

■

Theorem 1.32. *$(L^1[0,1], \|\cdot\|_1)$ is a Banach space.*

Proof. Let $\{f_n\}_{n\geq 1}$ be a Cauchy sequence in $(L^1[0,1], \|\cdot\|_1)$. Then for all $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that $\|f_n - f_m\|_1 < \varepsilon$ for all $n, m \geq N$. In particular, we can choose $\varepsilon = 2^{-k}$ for $k \in \mathbb{N}$. Then there exists $N_k \in \mathbb{N}$ such that $\|f_n - f_m\|_1 < 2^{-k}$ for all $n, m \geq N_k$. We can assume that $N_k < N_{k+1}$ for all $k \in \mathbb{N}$. Then define $g_k(x) = \sum_{i=1}^k \left|f_{N_{i+1}}(x) - f_{N_i}(x)\right|$. Then $\{g_k\}_{k\geq 1}$ is an increasing sequence of non-negative Lebesgue integrable functions, with $\int_0^1 g_k(x)\,dx < \sum_{i=1}^k 2^{-i} < 1$. By the monotone convergence theorem, there exists $g \in L^1[0,1]$ such that $g_k \uparrow g$ and $\int_0^1 g(x)\,dx = \lim_{k\to\infty} \int_0^1 g_k(x)\,dx < 1$. Now define $h_n(x) = f_n(x) - f_{N_1}(x)$ for all $n \geq N_1$. Then $|h_n(x)| = |f_n(x) - f_{N_1}(x)| \leq g(x)$ almost everywhere. By the dominated convergence theorem, there exists $h \in L^1[0,1]$ such that $h_n \to h$ almost everywhere and $\int_0^1 h(x)\,dx = \lim_{n\to\infty} \int_0^1 h_n(x)\,dx$. Finally, define $f = h + f_{N_1}$. Clearly, $f_n \to f$ almost everywhere, and $\int_0^1 f(x)\,dx = \lim_{n\to\infty} \int_0^1 f_n(x)\,dx$. Hence, $f \in L^1[0,1]$, and $\{f_n\}_{n\geq 1}$ converges to f in $(L^1[0,1], \|\cdot\|_1)$. Therefore, $(L^1[0,1], \|\cdot\|_1)$ is a Banach space. ■

The Fourier Transform

Let $f \in L^1[0,1]$. Then, for $n \in \mathbb{Z}$, we define the n-th *Fourier coefficient* of f as

$$c_n := \int_0^1 f(x) e^{-2\pi i n x}\,dx. \tag{1.41}$$

This integral is well-defined since

$$|c_n| = \left|\int_0^1 f(x) e^{-2\pi i n x}\,dx\right| \leq \int_0^1 |f(x)|\,dx = \|f\|_1 < \infty \tag{1.42}$$

(note here that we have not yet proved the middle inequality for complex-valued functions; we shall assume it for now). Thus, for a function $f \in L^1[0,1]$, we define the function $\hat{f} : \mathbb{Z} \to \mathbb{C}$ as $\hat{f}(n) = c_n$ for all $n \in \mathbb{Z}$. This function is called the *Fourier transform* of f.

August 20th.

A little on complex valued functions; the space $L^1([0,1],\mathbb{C})$ consists of all complex valued functions $f:[0,1]\to\mathbb{C}$ such that $\Re(f),\Im(f)\in L^1([0,1],\mathbb{R})$. Such an f is termed Lebesgue integrable, and the Lebesgue integral of f is defined as

$$\int_0^1 f\,dx := \int_0^1 \Re(f)\,dx + i\int_0^1 \Im(f)\,dx. \tag{1.43}$$

We show that all our previous results extend to complex valued functions.

Proposition 1.33. *Let $f=u+iv\in L^1([0,1],\mathbb{C})$, with $u,v\in L^1([0,1],\mathbb{R})$. Then $|f|\in L^1([0,1],\mathbb{R})$ and $\left|\int_0^1 f\,dx\right|\le\int_0^1|f|\,dx$.*

Proof. Note that $|f|=\sqrt{u^2+v^2}\le|u|+|v|$; there exist sequences of step functions $s_n\uparrow u$ and $t_n\uparrow v$ almost everywhere. Then $\sqrt{s_n^2+t_n^2}\to\sqrt{u^2+v^2}$ almost everywhere, and $\sqrt{s_n^2+t_n^2}\le\max|s_1|,|u|+\max|t_1|,|v|$ almost everywhere. By the dominated convergence theorem, $\int_0^1\sqrt{s_n^2+t_n^2}\,dx\to\int_0^1\sqrt{u^2+v^2}\,dx$. Hence, $|f|\in L^1([0,1],\mathbb{R})$. Now, $\int_0^1 f(x)\,dx=re^{i\theta}$ for some $r\ge0$ and $\theta\in[0,2\pi)$. Then $r=\int_0^1 e^{-i\theta}f(x)\,dx$ implies

$$\begin{aligned}\left|\int_0^1 f(x)\,dx\right| &= \left|\int_0^1 e^{-i\theta}f(x)\,dx\right| = \left|\int_0^1 \Re(e^{-i\theta}f(x))\,dx\right| \\ &\le \int_0^1\left|\Re(e^{-i\theta}f(x))\right|dx \le \int_0^1\left|e^{-i\theta}f(x)\right|dx = \int_0^1|f(x)|\,dx.\end{aligned} \tag{1.44}$$

■

Proposition 1.34 (The dominated convergence theorem for complex valued functions)**.** *Let $(f_n)_{n\ge1}\subseteq L^1([0,1],\mathbb{C})$ be a sequence of complex valued functions such that $f_n\to f$ almost everywhere, and there exists a Lebesgue integrable real function $g\ge0$ almost everywhere such that $|f_n|\le g$ almost everywhere for all $n\ge1$. Then $f\in L^1([0,1],\mathbb{C})$ and $\int_0^1 f\,dx=\lim_{n\to\infty}\int_0^1 f_n(x)\,dx$.*

Proof. If we write $f=u+iv$ and $f_n=u_n+iv_n$, with $u,v,u_n,v_n\in L^1([0,1],\mathbb{R})$, then $u_n\to u$ and $v_n\to v$ almost everywhere, and $|u_n|\le g$ and $|v_n|\le g$ almost everywhere for all $n\ge1$. Hence, by the dominated convergence theorem for real valued functions, we have $\int_0^1 u\,dx=\lim_{n\to\infty}\int_0^1 u_n(x)\,dx$ and $\int_0^1 v\,dx=\lim_{n\to\infty}\int_0^1 v_n(x)\,dx$. Thus, $\int_0^1 f\,dx=\lim_{n\to\infty}\int_0^1 f_n(x)\,dx$. ■

1.3.3 Integration on Unbounded Intervals

Simply put, such an integral can be viewed as the limit of integrals over bounded intervals. We shall work with the interval $\mathbb{R}$, but the same can be done for any unbounded interval. First off, the proto-monotone convergence theorem states that if $s_n\downarrow0$ almost everywhere, and with the additional condition that $\int_{\mathbb{R}}s_1(x)\,dx<\infty$, then $\int_{\mathbb{R}}s_n(x)\,dx\to0$. The set of upper functions $U(\mathbb{R})$ is defined as the upper limits of increasing sequences of *integrable* step functions. $L^1(\mathbb{R})$ is then defined as the span of $U(\mathbb{R})$, $\operatorname{span}U(\mathbb{R})$.

Theorem 1.35. *Let $f:[0,\infty)\to\mathbb{R}$, and suppose $f\chi_{[0,b]}\in L^1[0,b]$ for all $b>0$ and there exists $M\ge0$ such that $\int_0^b|f(x)|\,dx\le M$ for all $b>0$. Then*

1. *$f\in L^1[0,\infty)$,*
2. *$\lim_{b\to\infty}\int_0^b f(x)\,dx$ exists, and*
3. *$\int_0^\infty f(x)\,dx=\lim_{b\to\infty}\int_0^b f(x)\,dx$.*

Proof. Note that $|f|\,\chi_{[0,n]}\uparrow|f|$ almost everywhere, and $\int_0^\infty\left|f(x)\chi_{[0,n]}(x)\right|dx=\int_0^n|f(x)|\,dx\le M$ for all $n\ge1$. By the monotone convergence theorem, $\int_0^\infty|f(x)|\,dx=\lim_{n\to\infty}\int_0^n|f(x)|\,dx<\infty$. Hence, $|f|\in L^1[0,\infty)$. If we now let $g_n=f\chi_{[0,n]}$, then $g_n\to f$ almost everywhere, and $|g_n|\le|f|$ almost everywhere. By the dominated convergence theorem, $\int_0^\infty f(x)\,dx=\lim_{n\to\infty}\int_0^n f(x)\,dx$. ■

Recall the definition of the improper Riemann integral.

Definition 1.36. If f is Riemann integrable on $[0,b]$ for all $b>0$, and $\lim_{b\to\infty}(R)\int_0^b f(x)\,dx$ exists, then f is said to be *improperly Riemann integrable* on $[0,\infty)$, and the improper Riemann integral of f on $[0,\infty)$ is defined as $(IR)\int_0^\infty f(x)\,dx = \lim_{b\to\infty}(R)\int_0^b f(x)\,dx$.

Then,

Theorem 1.37. *Assume f is Riemann integrable on $[0,b]$ for all $b>0$, and that there exists a constant $M\geq 0$ such that $\left|\int_0^b f(x)\,dx\right| \leq M$ for all $b>0$. Then both f and $|f|$ are improperly Riemann integrable on $[0,\infty)$, and $f\in L^1[0,\infty)$, with $\int_0^\infty f(x)\,dx = (IR)\int_0^\infty f(x)\,dx$.*

Proof. Define $F(b)=\int_0^b |f(x)|\,dx$. Then F is increasing and bounded above by M, so $\lim_{b\to\infty}F(b)$ exists. Also, $0\leq |f(x)|-f(x)\leq 2|f(x)|$. By the dominated convergence theorem, $\lim_{b\to\infty}\int_0^b(|f(x)|-f(x))\,dx$ exists, and hence $\lim_{b\to\infty}\int_0^b |f(x)|\,dx$ exists. ■

Example 1.38. Let $f(x)=e^{-x}x^{t-1}$ for any $t>0$. We claim that $\int_0^\infty f(x)\,dx$ exists. Note that $\lim_{x\to\infty}x^n e^{-x}=0$ for all $n>0$. Hence, there exists $M_t\geq 0$ such that $e^{-x}x^{t-1}\leq e^{-x}M_t$ for all $x\geq 1$. Also, $\int_0^1 e^{-x}x^{t-1}\,dx\leq\int_0^1 x^{t-1}\,dx=\frac{1}{t}$, and $\int_1^b e^{-x}x^{t-1}\,dx\leq M_t\int_1^b e^{-x}\,dx = M_t(e^{-1}-e^{-b})$. Hence, $\left|\int_0^b f(x)\,dx\right|\leq\frac{1}{t}+M_t(e^{-1}-e^{-b})$, which is bounded above by $\frac{1}{t}+M_te^{-1}$ for all $b>0$. Hence, $\int_0^\infty f(x)\,dx$ exists.

In the above example, the function $\Gamma(t):=\int_0^\infty e^{-x}x^{t-1}\,dx$ is called the *Gamma function*. To find the derivative of Γ, we have

$$\Gamma'(t)=\lim_{n\to\infty}\frac{\Gamma(t+\frac{1}{n})-\Gamma(t)}{\frac{1}{n}} \tag{1.45}$$

By the definition of Γ,

$$\frac{\Gamma(t+\frac{1}{n})-\Gamma(t)}{\frac{1}{n}}=\int_0^\infty e^{-x}x^{t-1}\frac{x^{1/n}-1}{1/n}\,dx. \tag{1.46}$$

For $x>0$, the mean value theorem gives $\theta_{n,x}\in(0,1)$ such that

$$\frac{x^{1/n}-1}{1/n}=x^{\theta_{n,x}/n}\log x\longrightarrow\log x. \tag{1.47}$$

Choose N with $1/N<t/2$. For $n\geq N$, the integrand is dominated by

$$h(x)=x^{t-1-1/N}\,|\log x|\,\chi_{(0,1]}(x)+e^{-x}x^t\,|\log x|\,\chi_{[1,\infty)}(x), \tag{1.48}$$

which is integrable. Hence, by dominated convergence,

$$\Gamma'(t)=\int_0^\infty e^{-x}x^{t-1}\log x\,dx. \tag{1.49}$$

1.3.4 Measurable Functions

August 24th.

Definition 1.39. A function $f:[0,1]\to\mathbb{R}$ is said to be a *measurable function* if there exists a sequence of step functions $\{s_n\}_{n\geq 1}$ such that $s_n\to f$ almost everywhere. The set of all measurable functions on $[0,1]$ is denoted by $\mathcal{M}[0,1]$.

Notice that the definition of a measurable function is similar to that of an upper function, except that we do not require the step functions to be integrable. In fact, we claim that $L^1[0,1] \subsetneq \mathcal{M}[0,1]$. To see this, let $f \in L^1[0,1]$. Then there exists a sequence of step functions $\{s_n\}_{n\geq 1}$ such that $s_n \to f$ almost everywhere. Hence, f is measurable. However, the converse is not true; for example, the function $f(x) = \frac{1}{x}$ is measurable on $(0,1]$, but it is not Lebesgue integrable on $(0,1]$.

Theorem 1.40. *If $f \in \mathcal{M}[0,1]$ and there exists a non-negative $g \in L^1[0,1]$ such that $|f| \leq g$ almost everywhere, then $f \in L^1[0,1]$.*

Proof. Since f is measurable, there exists a sequence of step functions $\{s_n\}_{n\geq 1}$ such that $s_n \to f$ almost everywhere. Define $g_n = \max\{\min\{s_n, g\}, -g\}$ for all $n \geq 1$. Then $|g_n| \leq g$ almost everywhere, and $g_n \in L^1[0,1]$. Moreover, $g_n \to f$ almost everywhere. The dominated convergence theorem then implies that $f \in L^1[0,1]$. ■

Thus, to show a function f is integrable, it suffices to show that it is measurable and dominated by an integrable function.

Theorem 1.41. *Let $\varphi : \mathbb{R}^2 \to \mathbb{R}$ be continuous. If $f, g \in \mathcal{M}[0,1]$, then the function $h : [0,1] \to \mathbb{R}$ defined by $h(t) = \varphi(f(t), g(t))$ is also measurable. In particular, $f + g$, fg, $|f|$, $\max\{f,g\}$, and $\min\{f,g\}$ are all measurable. Also, $1/f \in \mathcal{M}[0,1]$ if f is non-zero almost everywhere.*

Proof. Note that if s and t are two step functions, then $\varphi(s,t)$ is also a step function; one can show this by considering the preimage of $\varphi(s,t)$ under the partition of $[0,1]$ induced by s and t. Since φ is continuous, whenever $s_n \to f$ and $t_n \to g$ almost everywhere, we have $\varphi(s_n, t_n) \to \varphi(f,g)$ almost everywhere. Hence, $\varphi(f,g)$ is measurable. The other statements follow from the fact that addition, multiplication, absolute value, maximum, minimum, and reciprocal (for non-zero functions) are all continuous operations. ■

Chapter 2

THE FOURIER TRANSFORM

Definition 2.1. For a function $f \in L^1(\mathbb{R}, \mathbb{C})$, the *Fourier transform* $\hat{f}$ of f is a complex-valued function on $\mathbb{R}$ defined by

$$\hat{f}(\xi) = \int_{-\infty}^{\infty} f(x) e^{-2\pi i x \xi} \, dx, \quad \xi \in \mathbb{R}. \tag{2.1}$$

One can show that $x \mapsto e^{-2\pi i x \xi}$ is measurable. Via this, the mapping $h(x) = f(x)e^{-2\pi i x \xi}$ is measurable, and dominated by $|f|$, which is integrable. Hence, the Fourier transform is well-defined. Also, we can show that it is continuous on $\mathbb{R}$:

$$\hat{f}(\xi + h_n) - \hat{f}(\xi) = \int_{-\infty}^{\infty} f(x) e^{-2\pi i x \xi} (e^{-2\pi i x h_n} - 1) \, dx \tag{2.2}$$

where the integrand coverges to 0 pointwise as $h_n \to 0$. Thus, by the dominated convergence theorem, we have $\hat{f}(\xi + h_n) \to \hat{f}(\xi)$ as $h_n \to 0$.

The notion of continuity may be redefined by our theory of Lebesgue integration.

Theorem 2.2. *Let X, Y be subintervals of $\mathbb{R}$, and $f : X \times Y \to \mathbb{R}$ be a function satisfying the following conditions.*

1. *For $y \in Y$, the function $f_y : X \to \mathbb{R}$ defined by $f_y(x) = f(x, y)$ is measurable.*
2. *There is a non0negative function $g \in L^1(X)$ such that $|f_y(x)| = |f(x, y)| \leq g(x)$ almost everywhere on X for all $y \in Y$.*
3. *For a fixed $y \in Y$, $\lim_{t \to y} f(x, t) = f(x, y)$ almost everywhere on X. That is, f_x is continuous almost everywhere.*

Then, $\int_X f(x, y) \, dx$ exists for each $y \in Y$, and the function $F : Y \to \mathbb{R}$ defined by $F(y) = \int_X f(x, y) \, dx$ is continuous on Y.

Proof. Note that $f_y \in \mathcal{M}(X)$ for each $y \in Y$, and $|f_y| \leq g$ almost everywhere on X. Hence, by the dominated convergence theorem, $f_y \in L^1(X)$. Let $y_n \to y$ in Y, and define $G_n(x) := f(x, y_n)$. Then $G_n(x) \to f(x, y)$ almost everywhere on X by the third assumption. Also, $F(y_n) = \int_X G_n(x) \, dx$ for each n. By the dominated convergence theorem, we have

$$\lim_{n \to \infty} F(y_n) = \lim_{n \to \infty} \int_X G_n(x) \, dx = \int_X f(x, y) \, dx = F(y). \tag{2.3}$$

■

Note that the Fourier transform $\hat{f}$ is continuous on $\mathbb{R}$ by the above theorem, but may fail to be differentiable.

Theorem 2.3. *Let X, Y be subintervals of $\mathbb{R}$, and $f : X \times Y \to \mathbb{R}$ be a function satisfying the following conditions.*

1. *For $y \in Y$, the function $f_y : X \to \mathbb{R}$ defined by $f_y(x) = f(x, y)$ is measurable, and $f_a \in L^1(X)$ for some $a \in Y$.*
2. *$\frac{\partial}{\partial y} f(x, y)$ exists for each interior point $(x, y) \in X \times Y$.*
3. *There exists a non-negative function $G \in L^1(X)$ such that $\left|\frac{\partial f}{\partial y}(x, y)\right| \leq G(x)$ for all interior points $(x, y) \in X \times Y$.*

Then $\int_X f(x, y)\, dx$ exists for all $y \in Y$, and the function $F : Y \to \mathbb{R}$ defined by $F(y) = \int_X f(x, y)\, dx$ is differentiable at each interior point of Y, and $F'(y) = \int_X \frac{\partial f}{\partial y}(x, y)\, dx$.

Proof. Since $f(x, y) - f(x, a) = (y - a)\frac{\partial}{\partial y} f(x, c)$ for some c between a and y, we have $|f_y(x)| \leq |f_a(x)| + |y - a|\, G(x)$, showing that $|f_y| \in L^1(X)$ for each $y \in Y$. Hence, $\int_X f(x, y)\, dx$ exists for all $y \in Y$. Let $y_n \to y$ in Y, and define $q_n(x) = \frac{f(x,y_n) - f(x,y)}{y_n - y}$. Then $q_n(x) \to \frac{\partial f}{\partial y}(x, y)$ for each interior point $(x, y) \in X \times Y$. Also, $|q_n(x)| \leq G(x)$ for all interior points $(x, y) \in X \times Y$. By the dominated convergence theorem, we have

$$\lim_{n\to\infty} \int_X q_n(x)\, dx = \int_X \frac{\partial f}{\partial y}(x, y)\, dx. \tag{2.4}$$

■

August 27th.

Example 2.4. Since $\chi_{[-1,1]} \in L^1(\mathbb{R})$, we can compute its Fourier transform:

$$\hat{\chi}_{[-1,1]}(\xi) = \int_{\mathbb{R}} \chi_{[-1,1]}(x) e^{-2\pi i x\xi}\, dx = \int_{-1}^{1} e^{-2\pi i x\xi}\, dx = \frac{e^{-2\pi i\xi} - e^{2\pi i\xi}}{-2\pi i\xi} = \frac{\sin(2\pi\xi)}{\pi\xi} = 2\,\mathrm{sinc}(2\pi\xi) \tag{2.5}$$

where $\mathrm{sinc}(x) = \frac{\sin x}{x}$ for $x \neq 0$, and $\mathrm{sinc}(0) = 1$.

For $y > 0$, define the function $F : (0, \infty) \to \mathbb{R}$ as

$$F(y) = \int_0^\infty e^{-xy}\, \mathrm{sinc}(x)\, dx. \tag{2.6}$$

Note that this integral is well-defined in the Lebesgue sense. Also, $|e^{-xy}\, \mathrm{sinc}(x)| \leq e^{-xy}$ for $x, y > 0$. We claim that F is continuous in its domain and vanishes at infinity. To this end, let $(y_n)_{n\geq 1}$ be a sequence of positive numbers such that $y_n \to y > 0$. Then

$$F(y_n) - F(y) = \int_0^\infty e^{-xy_n}(1 - e^{-x(y-y_n)})\, \mathrm{sinc}(x)\, dx \tag{2.7}$$

where $e^{-xy/2}(1 + e^{-xy}) \geq e^{-xy_n}(1 + e^{-xy_n})$ for sufficiently large n. Hence, by the dominated convergence theorem, we have $F(y_n) \to F(y)$ as $n \to \infty$. Also, for $y > 0$, we have

$$|F(y)| \leq \int_0^\infty e^{-xy}\, |\mathrm{sinc}(x)|\, dx \leq \int_0^\infty e^{-xy}\, dx = \frac{1}{y} \to 0 \quad \text{as } y \to \infty. \tag{2.8}$$

Now, $F'(y) = -\int_0^\infty e^{-xy} \sin(x)\, dx$ for $y > 0$. By integration by parts, and rearranging, we have

$$F'(y) = -\frac{1}{1 + y^2}. \tag{2.9}$$

Thus, $F(y) = C - \tan^{-1}(y)$ for some constant C. Since $F(y) \to 0$ as $y \to \infty$, we have $C = \frac{\pi}{2}$. Moreover, $\lim_{y\to 0^+} F(y) = \int_0^\infty \operatorname{sinc}(x)\,dx = \frac{\pi}{2}$. If we choose subintervals $X = [0,\infty)$ and $Y = [\varepsilon,\infty)$ for some $\varepsilon > 0$, then the function $f(x,y) = e^{-xy}\operatorname{sinc}(x)$ satisfies the hypotheses of the above theorem. Therefore, F is differentiable on $(0,\infty)$, and $F'(y) = -\int_0^\infty e^{-xy}\sin(x)\,dx$ for $y > 0$.

We also claim that $\lim_{\alpha\to\infty}\int_0^\alpha \operatorname{sinc}(x)\,dx = \frac{\pi}{2}$, as the improper Riemann integral. To see this, we know that

$$\lim_{\alpha\to\infty}\int_0^\alpha e^{-xy}\operatorname{sinc}(x)\,dx = \frac{\pi}{2} - \tan^{-1}(y) \quad \text{for } y > 0. \tag{2.10}$$

For $\varepsilon > 0$, choose $\beta > 0$ such that

$$\frac{5}{\beta} < \frac{\varepsilon}{3}. \tag{2.11}$$

By integration by parts, for all $\alpha > \beta$ and $y > 0$, we have

$$\left|\int_\beta^\alpha \operatorname{sinc}(x)\,dx\right| \le \frac{2}{\beta}, \tag{2.12}$$

$$\left|\int_\beta^\infty e^{-xy}\operatorname{sinc}(x)\,dx\right| \le \frac{3}{\beta}. \tag{2.13}$$

Choose $y > 0$ sufficiently small such that

$$\left|F(y) - \frac{\pi}{2}\right| < \frac{\varepsilon}{3} \quad \text{and} \quad \int_0^\beta (1 - e^{-xy})\,|\operatorname{sinc}(x)|\,dx < \frac{\varepsilon}{3}. \tag{2.14}$$

The second condition is possible since $e^{-xy} \to 1$ uniformly on $[0,\beta]$ as $y \to 0^+$. Hence, for all $\alpha > \beta$, we have

$$\begin{aligned}\left|\int_0^\alpha \operatorname{sinc}(x)\,dx - \frac{\pi}{2}\right| &\le \int_0^\beta (1 - e^{-xy})\,|\operatorname{sinc}(x)|\,dx + \left|\int_\beta^\alpha \operatorname{sinc}(x)\,dx\right| + \left|\int_\beta^\infty e^{-xy}\operatorname{sinc}(x)\,dx\right| + \left|F(y) - \frac{\pi}{2}\right| \\ &< \varepsilon.\end{aligned} \tag{2.15}$$

Therefore,

$$\lim_{\alpha\to\infty}\int_0^\alpha \operatorname{sinc}(x)\,dx = \frac{\pi}{2}. \tag{2.16}$$

2.1 Convolution

August 31st.

Definition 2.5. For $f, g \in L^1(\mathbb{R},\mathbb{C})$, the *convolution* $f * g$ of f and g is a complex-valued function on $\mathbb{R}$ defined by

$$(f * g)(x) = \int_{-\infty}^\infty f(x-t)g(t)\,dt, \quad x \in \mathbb{R}. \tag{2.17}$$

For example, if $g(x) = \frac{1}{2\varepsilon}\chi_{[-\varepsilon,\varepsilon]}(x)$ for some $\varepsilon > 0$, then

$$(f * g)(x) = \frac{1}{2\varepsilon}\int_{x-\varepsilon}^{x+\varepsilon} f(t)\,dt. \tag{2.18}$$

If f is continuous at x, then $(f * g)(x) \to f(x)$ as $\varepsilon \to 0^+$.

Proposition 2.6. *The following hold true for $f, g, h \in L^1(\mathbb{R},\mathbb{C})$.*

1. *Commutativity: $f * g = g * f$.*

2. *Associativity:* $f * (g * h) = (f * g) * h$.
3. *If g is smooth, then so is $f * g$.*
4. *If g is a polynomial on some interval, then so is $f * g$ on some interval.*

Also, we have the following result for the Fourier transform of a convolution.

Proposition 2.7. *For $f, g \in L^1(\mathbb{R}, \mathbb{C})$,*

$$\widehat{f * g} = \hat{f} \cdot \hat{g}. \tag{2.19}$$

Proof. Note that $f * g \in L^1(\mathbb{R}, \mathbb{C})$, and

$$\widehat{f * g}(\xi) = \int_{-\infty}^{\infty} (f * g)(x) e^{-2\pi i x \xi} \, dx = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x - t) g(t) e^{-2\pi i x \xi} \, dt \, dx. \tag{2.20}$$

By Fubini's theorem, we can interchange the order of integration:

$$\widehat{f * g}(\xi) = \int_{-\infty}^{\infty} g(t) \left(\int_{-\infty}^{\infty} f(x - t) e^{-2\pi i x \xi} \, dx \right) dt. \tag{2.21}$$

Let $u = x - t$, then $du = dx$, and we have

$$\widehat{f * g}(\xi) = \int_{-\infty}^{\infty} g(t) e^{-2\pi i t \xi} \left(\int_{-\infty}^{\infty} f(u) e^{-2\pi i u \xi} \, du \right) dt = \hat{f}(\xi) \cdot \hat{g}(\xi). \tag{2.22}$$

■

September 17th.

Theorem 2.8. *Let X, Y be subintervals of $\mathbb{R}$, and suppose $k : X \times Y \to \mathbb{R}$ is continuous and bounded. With $f \in L^1(X)$ and $g \in L^1(Y)$, the following hold true.*

1. *For $y \in Y$, $\int_X f(x)k(x, y)dx$ exists and $F(y) = \int_X f(x)k(x, y)dx$ is continuous on Y.*
2. *For $x \in X$, $\int_Y g(y)k(x, y)dy$ exists and $G(x) = \int_Y g(y)k(x, y)dy$ is continuous on X.*
3. *The integrals $\int_X f(x)G(x)dx$ and $\int_Y g(y)F(y)dy$ exist, and*

$$\int_Y g(y) \left(\int_X f(x)k(x, y) \, dx \right) dy = \int_Y g(y)F(y) \, dy = \int_X f(x)G(x) \, dx = \int_X f(x) \left(\int_Y g(y)k(x, y) \, dy \right) dx. \tag{2.23}$$

Proof. Let $M = \sup_{X \times Y} |k| < \infty$.

For the first statement, fix $y \in Y$ and let $f_y(x) = f(x)k(x, y)$. Since f is measurable and $x \mapsto k(x, y)$ is continuous, f_y is measurable, and $|f_y(x)| \leq M |f(x)|$ with $M |f| \in L^1(X)$. By the dominated convergence theorem, $f_y \in L^1(X)$, so $F(y) = \int_X f(x)k(x, y) \, dx$ exists. Now let $y_n \to y$ in Y. Then

$$|F(y_n) - F(y)| \leq \int_X |f(x)| \, |k(x, y_n) - k(x, y)| \, dx \tag{2.24}$$

where the integrand converges to 0 pointwise as $n \to \infty$ by the continuity of k, and is dominated by $2M |f(x)| \in L^1(X)$. Hence, by the dominated convergence theorem, we have $F(y_n) \to F(y)$ as $n \to \infty$, so F is continuous on Y. The second statement follows by interchanging the roles of (X, f) and (Y, g).

For the third statement, note that G is continuous by the second statement, with $|G(x)| \leq M \|g\|_1$. Thus fG is measurable and $|f(x)G(x)| \leq M \|g\|_1 |f(x)|$, so $fG \in L^1(X)$; likewise $|g(y)F(y)| \leq M \|f\|_1 |g(y)|$, so $gF \in L^1(Y)$. Both integrals therefore exist, and it remains to prove that $\int_X f(x)G(x) \, dx = \int_Y g(y)F(y) \, dy$. For $\phi \in L^1(X)$ and $\psi \in L^1(Y)$, write

$$I(\phi, \psi) = \int_X \phi(x) \left(\int_Y \psi(y)k(x, y) \, dy \right) dx, \qquad J(\phi, \psi) = \int_Y \psi(y) \left(\int_X \phi(x)k(x, y) \, dx \right) dy, \tag{2.25}$$

so that the desired identity is $I(f,g) = J(f,g)$. Both I and J are bilinear, and since the inner integrals are bounded by $M\|\psi\|_1$ and $M\|\phi\|_1$ respectively,

$$|I(\phi,\psi)| \le M\|\phi\|_1\|\psi\|_1, \qquad |J(\phi,\psi)| \le M\|\phi\|_1\|\psi\|_1. \tag{2.26}$$

If $\phi = \chi_A$ and $\psi = \chi_B$ for compact intervals $A \subseteq X$ and $B \subseteq Y$, then k is continuous on the compact rectangle $A \times B$, and the theory of the multivariable Riemann integral gives

$$I(\chi_A,\chi_B) = \int_A \left(\int_B k(x,y)\,dy\right)dx = \int_B \left(\int_A k(x,y)\,dx\right)dy = J(\chi_A,\chi_B). \tag{2.27}$$

By bilinearity, $I(s,t) = J(s,t)$ for all step functions s on X and t on Y.

Now let $\varepsilon > 0$, and choose step functions s and t with $\int_X |f-s|\,dx < \varepsilon$ and $\int_Y |g-t|\,dy < \varepsilon$, so that $\|s\|_1 \le \|f\|_1 + \varepsilon$. Using bilinearity together with the bounds above,

$$|I(f,g) - I(s,t)| = |I(f-s,g) + I(s,g-t)| \le M\|g\|_1\varepsilon + M(\|f\|_1+\varepsilon)\varepsilon, \tag{2.28}$$

and identically $|J(f,g) - J(s,t)| \le M\|g\|_1\varepsilon + M(\|f\|_1+\varepsilon)\varepsilon$. Since $I(s,t) = J(s,t)$, it follows that

$$|I(f,g) - J(f,g)| \le |I(f,g) - I(s,t)| + |J(s,t) - J(f,g)| \le 2M\varepsilon\left(\|f\|_1 + \|g\|_1 + \varepsilon\right). \tag{2.29}$$

Letting $\varepsilon \to 0^+$ yields $I(f,g) = J(f,g)$, as required. ■

We show more properties of the convolution.

Proposition 2.9. *Let $f, g \in L^1(\mathbb{R})$.*

1. *If $f, g \in C_c(\mathbb{R})$, then $f * g \in C_c(\mathbb{R})$.*
2. *If $f \in L^1(\mathbb{R})$ and $g \in C_c(\mathbb{R})$, then $f * g \in C(\mathbb{R})$.*
3. *Let $\tau_p f(x) = f(x-p)$ for $p \in \mathbb{R}$. Then $\tau_p(f*g) = (\tau_p f) * g = f * (\tau_p g)$.*
4. *$(f*g)' = f' * g = f * g'$.*
5. *Let $\varphi \in C_c^\infty(\mathbb{R})$ be such that $\int_{\mathbb{R}} \varphi(x)dx = 1$, and φ is an even non-negative function that is monotonically decreasing on $[0,\infty)$. Then, with $\varphi_\varepsilon(x) = \frac{1}{\varepsilon}\varphi(\frac{x}{\varepsilon})$, we have $\lim_{\varepsilon\to 0^+} f * \varphi_\varepsilon = f$ in $L^1(\mathbb{R})$.*
6. *If, additionally, $f \in C_c(\mathbb{R})$, then $\lim_{\varepsilon\to 0^+} f * \varphi_\varepsilon = f$ uniformly on $\mathbb{R}$.*

Proof. 1. Since the integrand of $(f*g)(x) = \int_{\mathbb{R}} f(x-y)g(y)\,dy$ is nonzero only when $y \in \operatorname{supp}(g)$ and $x - y \in \operatorname{supp}(f)$, any x with $(f*g)(x) \neq 0$ satisfies $x = (x-y) + y \in \operatorname{supp}(f) + \operatorname{supp}(g)$. As $\operatorname{supp}(f)$ and $\operatorname{supp}(g)$ are compact, so is their sum $\operatorname{supp}(f) + \operatorname{supp}(g)$, being the image of the compact set $\operatorname{supp}(f) \times \operatorname{supp}(g)$ under addition; hence $\operatorname{supp}(f*g) \subseteq \operatorname{supp}(f) + \operatorname{supp}(g)$ is compact. Moreover $f \in C_c(\mathbb{R}) \subseteq L^1(\mathbb{R})$ and $g \in C_c(\mathbb{R})$, so $f*g$ is continuous by (2). Therefore $f * g \in C_c(\mathbb{R})$.

2. Since $g \in C_c(\mathbb{R})$ is bounded and uniformly continuous, commutativity gives $(f*g)(x) = \int_{\mathbb{R}} f(y)g(x-y)\,dy$, so for $h \in \mathbb{R}$,

$$|(f*g)(x+h) - (f*g)(x)| \le \int_{\mathbb{R}} |f(y)|\,|g(x+h-y) - g(x-y)|\,dy \le \|f\|_1 \sup_{z\in\mathbb{R}} |g(z+h) - g(z)|. \tag{2.30}$$

By the uniform continuity of g, the supremum tends to 0 as $h \to 0$, uniformly in x. Hence $f*g$ is (uniformly) continuous on $\mathbb{R}$.

3. Directly, and using the substitution $u = y - p$ where needed,

$$(\tau_p f * g)(x) = \int_{\mathbb{R}} f(x-y-p)g(y)\,dy = (f*g)(x-p) = \tau_p(f*g)(x), \tag{2.31}$$

$$(f * \tau_p g)(x) = \int_{\mathbb{R}} f(x-y)g(y-p)\,dy = \int_{\mathbb{R}} f(x-p-u)g(u)\,du = (f*g)(x-p). \tag{2.32}$$

Hence $\tau_p(f*g) = (\tau_p f) * g = f * (\tau_p g)$.

4. Suppose $g \in C_c^1(\mathbb{R})$. Writing $(f*g)(x) = \int_{\mathbb{R}} f(y)g(x-y)\,dy$, the integrand has x-derivative $f(y)g'(x-y)$, which is dominated by $\|g'\|_\infty |f(y)| \in L^1(\mathbb{R})$. By differentiation under the integral sign,

$$(f*g)'(x) = \int_{\mathbb{R}} f(y)g'(x-y)\,dy = (f*g')(x). \tag{2.33}$$

Symmetrically, if $f \in C_c^1(\mathbb{R})$, then writing $(f*g)(x) = \int_{\mathbb{R}} f(x-y)g(y)\,dy$ and differentiating under the integral sign gives $(f*g)'(x) = \int_{\mathbb{R}} f'(x-y)g(y)\,dy = (f'*g)(x)$. In particular, when both derivatives exist, $(f*g)' = f'*g = f*g'$.

5. Since $\int_{\mathbb{R}} \varphi_\varepsilon(y)\,dy = \int_{\mathbb{R}} \varphi(y)\,dy = 1$ for every $\varepsilon > 0$, we may write

$$(f*\varphi_\varepsilon)(x) - f(x) = \int_{\mathbb{R}} (f(x-y) - f(x))\,\varphi_\varepsilon(y)\,dy. \tag{2.34}$$

Taking the L^1-norm and using Tonelli's theorem to interchange the integrals,

$$\|f*\varphi_\varepsilon - f\|_1 \le \int_{\mathbb{R}} \varphi_\varepsilon(y) \left(\int_{\mathbb{R}} |f(x-y) - f(x)|\,dx \right) dy = \int_{\mathbb{R}} \varphi_\varepsilon(y)\,\omega(y)\,dy, \tag{2.35}$$

where $\omega(y) = \|\tau_y f - f\|_1$. Recall that translation is continuous in $L^1(\mathbb{R})$, that is, $\omega(y) \to 0$ as $y \to 0$: given $\eta > 0$, choose $\psi \in C_c(\mathbb{R})$ with $\|f-\psi\|_1 < \eta$, so that

$$\omega(y) \le \|\tau_y f - \tau_y\psi\|_1 + \|\tau_y\psi - \psi\|_1 + \|\psi - f\|_1 = 2\,\|f-\psi\|_1 + \|\tau_y\psi - \psi\|_1\,, \tag{2.36}$$

and $\|\tau_y\psi - \psi\|_1 \to 0$ as $y \to 0$ by the uniform continuity of ψ; also $\omega(y) \le 2\,\|f\|_1$ for all y. Now fix $\eta > 0$ and choose $\delta > 0$ with $\omega(y) < \eta$ for $|y| < \delta$. Then

$$\|f*\varphi_\varepsilon - f\|_1 \le \int_{|y|<\delta} \varphi_\varepsilon(y)\,\omega(y)\,dy + \int_{|y|\ge\delta} \varphi_\varepsilon(y)\,\omega(y)\,dy \le \eta + 2\,\|f\|_1 \int_{|y|\ge\delta} \varphi_\varepsilon(y)\,dy. \tag{2.37}$$

Finally, $\int_{|y|\ge\delta} \varphi_\varepsilon(y)\,dy = \int_{|u|\ge\delta/\varepsilon} \varphi(u)\,du \to 0$ as $\varepsilon \to 0^+$, since $\varphi \in L^1(\mathbb{R})$. Hence we have the bound $\limsup_{\varepsilon\to 0^+} \|f*\varphi_\varepsilon - f\|_1 \le \eta$, and as $\eta > 0$ was arbitrary, $\lim_{\varepsilon\to 0^+} f*\varphi_\varepsilon = f$ in $L^1(\mathbb{R})$. ∎

September 21st.

Theorem 2.10. *If $f, g \in L^1(\mathbb{R})$, then $f*g \in L^1(\mathbb{R})$, and $\|f*g\|_1 \le \|f\|_1\,\|g\|_1$. Further, if $f, g \ge 0$, then $\|f*g\|_1 = \|f\|_1\,\|g\|_1$.*

Proof. First, we show for indicator functions χ_I and χ_J. We have

$$(\chi_I * \chi_J)(x) = \int_{\mathbb{R}} \chi_I(x-y)\chi_J(y)\,dy = \int_{\mathbb{R}} \chi_{(x-I)\cap J}(y)\,dy = |(x-I)\cap J|\,, \tag{2.38}$$

so that

$$\|\chi_I * \chi_J\|_1 = \int_{\mathbb{R}} |(x-I)\cap J|\,dx = \int_{\mathbb{R}}\int_{\mathbb{R}} \chi_{(x-I)\cap J}(y)\,dy\,dx = \int_{\mathbb{R}}\int_{\mathbb{R}} \chi_I(x-y)\chi_J(y)\,dx\,dy = |I|\,|J|\,. \tag{2.39}$$

Now let $s = \sum_{i=1}^p c_i\chi_{I_i}$ and $t = \sum_{j=1}^q d_j\chi_{J_j}$ be step functions. Then

$$(s*t)(x) = \sum_{i,j} c_i d_j\,|(x-I_i)\cap J_j| \implies \|s*t\|_1 = \sum_{i,j} c_i d_j\,|I_i|\,|J_j| \le \|s\|_1\,\|t\|_1\,. \tag{2.40}$$

For general $f, g \in L^1(\mathbb{R})$, we can approximate them by step functions; let $f, g \ge 0$ with $f, g \in L^1(\mathbb{R})$. Choose step functions $s_n \uparrow f$ and $t_n \uparrow g$ pointwise almost everywhere. Then define

$$\psi_n(x) := (s_n * t_n)(x) = \int_{\mathbb{R}} s_n(x-y)t_n(y)\,dy. \tag{2.41}$$

Since $s_n \uparrow f$, it converges outside a null set N_f. Thus, $(y \mapsto s_n(x-y)) \uparrow (y \mapsto f(x-y))$ outside $x - N_f$. Similarly, $(y \mapsto t_n(y)) \uparrow (y \mapsto g(y))$ outside N_g. So,

$$\left|\int_{\mathbb{R}} \psi_n(x)\,dx\right| = \left|\int_{\mathbb{R}} s_n(x)\,dx\right| \left|\int_{\mathbb{R}} t_n(x)\,dx\right| \leq \|s_n\|_1 \|t_n\|_1 \leq \|f\|_1 \|g\|_1 . \tag{2.42}$$

Thus, by the monotone convergence theorem, we have that $\psi := \lim_{n\to\infty} \psi_n$ exists with

$$\int_{\mathbb{R}} \psi(x)\,dx = \lim_{n\to\infty} \int_{\mathbb{R}} \psi_n(x)\,dx = \lim_{n\to\infty} \|s_n\|_1 \|t_n\|_1 = \|f\|_1 \|g\|_1 . \tag{2.43}$$

It remains to show that $\psi = f * g$ almost everywhere. For $x \in \mathbb{R}$, we have

$$\psi(x) = \lim_{n\to\infty} \psi_n(x) = \lim_{n\to\infty} \int_{\mathbb{R}} s_n(x-y)t_n(y)\,dy = \int_{\mathbb{R}} f(x-y)g(y)\,dy = (f*g)(x). \tag{2.44}$$

Therefore, $\psi = f * g$ almost everywhere. Finally, for general $f, g \in L^1(\mathbb{R})$, we can write $f = f_+ - f_-$ and $g = g_+ - g_-$, where $f_\pm, g_\pm \geq 0$ are in $L^1(\mathbb{R})$. Then

$$f * g = (f_+ - f_-) * (g_+ - g_-) = f_+ * g_+ - f_+ * g_- - f_- * g_+ + f_- * g_-, \tag{2.45}$$

and the result follows. ■

Theorem 2.11 (The *Weierstrass approximation theorem*)**.** *The space of polynomial functions on* $[0,1]$ *is dense in* $(C[0,1], \|\cdot\|_\infty)$.

Proof. Let $f \in C[0,1]$. Without the loss of generality, we may assume $f(0) = f(1) = 0$, since we can set $g(x) = f(x) - f(0) - x(f(1) - f(0))$ and approximate g instead. Extend f to a continuous function on $\mathbb{R}$ by setting $f(x) = 0$ for $x \notin [0,1]$, so that $f \in C_c(\mathbb{R}) \subseteq L^1(\mathbb{R})$. Let Q_n be the function defined as $Q_n(x) = c_n(1-x^2)^n$ for $x \in [-1,1]$ and $Q_n(x) = 0$ for $x \notin [-1,1]$, where c_n is a normalization constant. Then $Q_n \in C_c(\mathbb{R})$ is non-negative, even, and monotonically decreasing on $[0,\infty)$, with $\int_{\mathbb{R}} Q_n(x)\,dx = 1$. Define $P_n(x) = f * Q_n(x) = \int_0^1 f(t)Q_n(t-x)\,dt$ for $x \in [0,1]$. ■

(INCOMPLETE)

2.2 Algebras of Functions

Definition 2.12. Let E be a set. A collection of functions $\mathcal{A} \subseteq \mathbb{C}^E$ is said to be an *algebra* if the following hold.

1. $f + g \in \mathcal{A}$ for all $f, g \in \mathcal{A}$.
2. $fg \in \mathcal{A}$ for all $f, g \in \mathcal{A}$.
3. $\lambda f \in \mathcal{A}$ for all $f \in \mathcal{A}$ and $\lambda \in \mathbb{C}$.

$\mathcal{A}$ is said to be *uniformly closed* if $(\mathcal{A}, \|\cdot\|_\infty)$ is complete.

Examples include the set of polynomials on $[0,1]$, the set of trigonometric polynomials on $[0, 2\pi]$, and the set of continuous functions on a compact interval $[a,b]$.

Definition 2.13. Let $\mathcal{A} \subseteq \mathbb{C}^E$ be an algebra.

1. $\mathcal{A}$ is said to separate points on E if for any $x, y \in E$ with $x \neq y$, there exists $f \in \mathcal{A}$ such that $f(x) \neq f(y)$.
2. For each $x \in E$, there exists $g \in \mathcal{A}$ such that $g(x) \neq 0$; in this case, we say that $\mathcal{A}$ vanishes nowhere on E.

September 24th.

Theorem 2.14. *Let $\mathcal{A} \subseteq \mathbb{C}^E$ be an algebra over E, with $|E| \geq 2$, such that $\mathcal{A}$ separate points on E and vanishes nowhere on E. Take $x_1 \neq x_2$ on E, and $c_1, c_2 \in \mathbb{C}$. Then there exists $f \in \mathcal{A}$ such that $f(x_1) = c_1$ and $f(x_2) = c_2$.*

Proof. Note that there exist $g, h, k \in \mathcal{A}$ such that $g(x_1) \neq g(x_2)$, $h(x_1) \neq 0$, and $k(x_2) \neq 0$. Then set $u = gk - g(x_1)k$ and $v = gh - g(x_2)h$, so that $u, v \in \mathcal{A}$. We have $u(x_1) = v(x_2) = 0$, and $u(x_2) \neq 0$ and $v(x_1) \neq 0$. Now define

$$f = c_1 \frac{v}{v(x_1)} + c_2 \frac{u}{u(x_2)} \in \mathcal{A}. \tag{2.46}$$

Then $f(x_1) = c_1$ and $f(x_2) = c_2$. ■

Theorem 2.15 (The *discrete Stone-Weierstrass theorem*)**.** *Let $\mathcal{A} \subseteq \mathbb{R}^E$ be an algebra of real-valued functions on a finite set $E = \{x_1, \ldots, x_n\}$. If $\mathcal{A}$ separates points on E and vanishes nowhere on E, then $\mathcal{A} = \mathbb{R}^E$.*

Proof. Let $f \in \mathbb{R}^E$. For each $i = 1, \ldots, n$, there exists $g_i \in \mathcal{A}$ such that $g_i(x_i) = f(x_i)$ and $g_i(x_j) = 0$ for all $j \neq i$. Then $f = g_1 + \cdots + g_n \in \mathcal{A}$. ■

Theorem 2.16 (The *Stone-Weierstrass theorem*)**.** *Let $\mathcal{A} \subseteq C(K)$ be an algebra of real-valued continuous functions on a compact Hausdorff space K. If $\mathcal{A}$ separates points on K and vanishes nowhere on K, then $\mathcal{A}$ is dense in $(C(K), \|\cdot\|_\infty)$.*

Proof. We show that $\overline{\mathcal{A}} = C(K)$, where $\overline{\mathcal{A}}$ is the uniform closure of $\mathcal{A}$. Note that $\overline{\mathcal{A}}$ is an algebra, separates points on K, and vanishes nowhere on K. If $f \in \overline{\mathcal{A}}$, then $|f| \in \overline{\mathcal{A}}$; consider the map $x \mapsto |x|$ on $[-a, a]$ for $a > 0$. By the Weierstrass approximation theorem, there is a sequence of polynomial functions p_n on $[-a, a]$ such that $p_n(x) \to |x|$ on $[-a, a]$ uniformly. For every $\varepsilon > 0$, there exists a polynomial function p on $[-a, a]$ such that $|p(x) - |x|| < \varepsilon$ for all $x \in [-a, a]$. Thus, set $g = p(f) \in \overline{\mathcal{A}}$, and $\|g - |f|\|_\infty < \varepsilon$. Hence $|f| \in \overline{\mathcal{A}}$.

By the above, if f, g are in $\overline{\mathcal{A}}$, then so are $\max\{f, g\}$ and $\min\{f, g\}$. Indeed, $\max\{f, g\} = \frac{1}{2}(f + g + |f - g|)$ and $\min\{f, g\} = \frac{1}{2}(f + g - |f - g|)$. We claim that if $f \in C(K)$, $x \in K$, and $\varepsilon > 0$, then there exists $g_x \in \overline{\mathcal{A}}$ such that $g_x(x) = f(x)$, and $g_x(t) > f(t) - \varepsilon$ for all $t \in K$; to this end, let $y \in K$ with $y \neq x$. By the hypothesis, there exists $h_y \in \overline{\mathcal{A}}$ such that $h_y(x) = f(x)$ and $h_y(y) = f(y)$. By continuity, there is a neighbourhood J_y of x such that $h_y(t) > f(t) - \varepsilon$ for all $t \in J_y$. Since y was arbitrary, the collection $\{J_y\}_{y \in K \setminus \{x\}}$ is an open cover of K. K being compact, there exists a finite subcover $\{J_{y_1}, \ldots, J_{y_n}\}$, and we can set $g_x = \max\{h_{y_1}, \ldots, h_{y_n}\} \in \overline{\mathcal{A}}$. On K, we have $g_x(t) > f(t) - \varepsilon$ for all $t \in K$, and $g_x(x) = f(x)$.

Via the above, we can show that given $f \in C(K)$ and $\varepsilon > 0$, there exists $h \in \overline{\mathcal{A}}$ such that $\|f - h\|_\infty < \varepsilon$. Indeed, for $x \in K$, there exists $g_x \in \overline{\mathcal{A}}$ such that $g_x(x) = f(x)$ and $g_x(t) > f(t) - \varepsilon$ for all $t \in K$. By continuity, again, there is a neighbourhood N_x of x such that $g_x(t) < f(t) + \varepsilon$ for all $t \in N_x$. The collection $\{N_x\}_{x \in K}$ is an open cover of K, and by compactness, there exists a finite subcover $\{N_{x_1}, \ldots, N_{x_m}\}$. Set $h = \min\{g_{x_1}, \ldots, g_{x_m}\} \in \overline{\mathcal{A}}$. Then for all $t \in K$, we have $f(t) - \varepsilon < h(t) < f(t) + \varepsilon$, so that $\|f - h\|_\infty < \varepsilon$. Therefore $\overline{\mathcal{A}} = C(K)$. ■

An algebra $\mathcal{A} \subseteq C(K; \mathbb{C})$ is said to be *self-adjoint* if $f \in \mathcal{A}$ implies $\overline{f} \in \mathcal{A}$, where $\overline{f}(x) = \overline{f(x)}$ for all $x \in K$. The following is a generalization of the Stone-Weierstrass theorem to complex-valued functions.

Theorem 2.17. *Suppose $\mathcal{A} \subseteq C(K; \mathbb{C})$ is a self-adjoint algebra of complex-valued continuous functions on a compact Hausdorff space K, such that $\mathcal{A}$ separates points on K and vanishes nowhere on K. Then $\mathcal{A}$ is dense in $(C(K; \mathbb{C}), \|\cdot\|_\infty)$.*

Proof. Let $\mathcal{A}_\mathbb{R} = \{\Re(f), \Im(f) \mid f \in \mathcal{A}\}$, the real and imaginary parts of functions in $\mathcal{A}$. Let $h, k \in \mathcal{A}_\mathbb{R}$ be such that $h = \Re(f)$ and $k = \Im(g) = \Re(-ig)$ for some $f, g \in \mathcal{A}$. Then $h + k = \Re(f - ig) \in \mathcal{A}_\mathbb{R}$, and $\lambda h = \Re(\lambda f) \in \mathcal{A}_\mathbb{R}$ for $\lambda \in \mathbb{R}$. Also, $hk = \frac{1}{4}(|f + g|^2 - |f - g|^2) \in \mathcal{A}_\mathbb{R}$, since $|f|^2 = f\overline{f} \in \mathcal{A}$ by self-adjointness. Hence $\mathcal{A}_\mathbb{R}$ is an algebra of real-valued continuous functions on K. It separates points on K and vanishes nowhere on K, since if $x, y \in K$ with $x \neq y$, there exists $f \in \mathcal{A}$ such that $f(x) \neq f(y)$, so

that either $\Re(f)(x) \neq \Re(f)(y)$ or $\Im(f)(x) \neq \Im(f)(y)$; also, for each $x \in K$, there exists $f \in \mathcal{A}$ such that $f(x) \neq 0$, so that either $\Re(f)(x) \neq 0$ or $\Im(f)(x) \neq 0$. By the Stone-Weierstrass theorem, $\mathcal{A}_{\mathbb{R}}$ is dense in $(C(K;\mathbb{R}), \|\cdot\|_\infty)$. Since every function in $\mathcal{A}$ can be written as a linear combination of two functions in $\mathcal{A}_{\mathbb{R}}$, it follows that $\mathcal{A}$ is dense in $(C(K;\mathbb{C}), \|\cdot\|_\infty)$. ■

Example 2.18. Let $K = [0,1]$, and let $\mathcal{A}$ be the set

$$\left\{ x \mapsto \sum_{n=-N}^{N} a_n e^{2\pi i n x} \;\middle|\; N \in \mathbb{N}_0,\ a_n \in \mathbb{C} \right\}. \tag{2.47}$$

$\mathcal{A}$, here, is termed the set of *trigonometric polynomials*. Note that $f(0) = f(1)$ for all $f \in \mathcal{A}$, so $\mathcal{A}$ does not separate points on K. However, if we identify 0 and 1, then $\mathcal{A}$ separates points on the resulting quotient space, which is homeomorphic to the unit circle. Then $\mathcal{A}$ is a self-adjoint algebra of complex-valued continuous functions on the unit circle that separates points and vanishes nowhere, so by the Stone-Weierstrass theorem, $\mathcal{A}$ is dense in $(C(S^1;\mathbb{C}), \|\cdot\|_\infty)$.

2.3 Fourier Series

September 28th.

Let $f \in L^1(S^1)$, or $f \in L^1[0,1]$ with $f(0) = f(1)$. Recall that the n-th Fourier coefficient of f is defined as

$$\hat{f}(n) = \int_0^1 f(x) e^{-2\pi i n x}\, dx, \qquad n \in \mathbb{Z}. \tag{2.48}$$

If we identify $c_n \equiv c_n(f) = \hat{f}(n)$, then we have the map

$$L^1(S^1) \to \mathbb{C}^{\mathbb{Z}}, \qquad f \mapsto (\ldots, c_{-2}, c_{-1}, c_0, c_1, c_2, \ldots), \tag{2.49}$$

for any $f \in L^1(S^1)$. The *Fourier series* of f, then, is the formal series

$$x \mapsto \sum_{n\in\mathbb{Z}} c_n(f) e^{2\pi i n x}, \qquad x \in [0,1]. \tag{2.50}$$

In particular, let us find the Fourier coefficients of $f(x) = e^{2\pi i m x}$ for $m \in \mathbb{Z}$. We have

$$c_n(f) = \int_0^1 e^{2\pi i m x} e^{-2\pi i n x}\, dx = \int_0^1 e^{2\pi i (m-n) x}\, dx = \begin{cases} 1, & m = n, \\ 0, & m \neq n \end{cases} = \delta_{mn}. \tag{2.51}$$

That is, the Fourier series of f is simply f itself. More generally, if we take finite linear combinations of such functions, we obtain the set of trigonometric polynomials, and the Fourier series of a trigonometric polynomial is simply the polynomial itself.

Lemma 2.19 (The *Riemann-Lebesgue lemma*)**.** *If $f \in L^1(S^1)$, then $\lim_{n\to\infty} c_n(f) = 0$ and $\lim_{n\to-\infty} c_n(f) = 0$.*

Proof. Let $I = [a,b]$ be a subinterval of $[0,1]$, and let $f = \chi_I$ be the indicator function of I. Then

$$c_n(\chi_I) = \int_0^1 \chi_I(x) e^{-2\pi i n x}\, dx = \int_a^b e^{-2\pi i n x}\, dx = \frac{e^{-2\pi i n b} - e^{-2\pi i n a}}{-2\pi i n} \to 0 \quad \text{as } n \to \pm\infty. \tag{2.52}$$

The result also follows for simple functions. For general $f \in L^1[0,1]$, let $(f_k)_{k\geq 1}$ be a sequence of step functions such that $f_k \to f$ in $L^1[0,1]$. Then

$$|c_n(f)| \leq |c_n(f) - c_n(f_k)| + |c_n(f_k)| \leq \|f - f_k\|_1 + |c_n(f_k)|. \tag{2.53}$$

Now, for $\varepsilon > 0$, choose k large enough that $\|f - f_k\|_1 < \varepsilon/2$. Then choose N large enough that $|c_n(f_k)| < \varepsilon/2$ for all $|n| > N$. Hence, for all $|n| > N$, we have $|c_n(f)| < \varepsilon$, and the result follows. ■

Therefore, the map $L^1(S^1) \to \mathbb{C}^{\mathbb{Z}}$, $f \mapsto (c_n(f))_{n\in\mathbb{Z}}$, is a bounded linear map, actually lands in $\ell^\infty(\mathbb{Z}) \subseteq \mathbb{C}^{\mathbb{Z}}$, and is continuous with respect to the L^1-norm on $L^1(S^1)$ and the ℓ^∞-norm on $\ell^\infty(\mathbb{Z})$. We provide another proof of this lemma for continuous functions, while making use of the *modulus of continuity* of a function. For $f \in C(S^1)$, define the modulus of continuity $\omega_f : [0,1] \to [0,\infty)$ by

$$\omega_f(\delta) = \sup_{|h|\le\delta} \sup_{\theta} |f(\theta+h) - f(\theta)|. \tag{2.54}$$

That is, we are looking at the maximum perturbation of f over all intervals of length δ.

Theorem 2.20. *Let $f \in C(S^1)$. Then*

$$\left|\hat{f}(n)\right| \le \frac{1}{2}\omega_f\left(\frac{1}{2|n|}\right), \tag{2.55}$$

implying $\hat{f}(n) \to 0$ as $|n| \to \infty$.

Proof. Let $h = 1/2n$. Then

$$\int_0^1 f(\theta)e^{-2\pi in\theta}\,d\theta = \hat{f}(n) = \int_0^1 f(\theta+h)e^{-2\pi in(\theta+h)}\,d\theta = -\int_0^1 f(\theta+h)e^{-2\pi in\theta}\,d\theta. \tag{2.56}$$

Adding these two integrals, we get

$$\hat{f}(n) = \frac{1}{2}\int_0^1 (f(\theta) - f(\theta+h))\,e^{-2\pi in\theta}\,d\theta, \tag{2.57}$$

or

$$\left|\hat{f}(n)\right| \le \frac{1}{2}\int_0^1 |f(\theta) - f(\theta+h)|\,d\theta \le \frac{1}{2}\omega_f(h) = \frac{1}{2}\omega_f\left(\frac{1}{2|n|}\right). \tag{2.58}$$

Since $f \in C(S^1)$, $\omega(\delta) \to 0$ as $\delta \to 0^+$, and the result follows. ∎

The same style of proof may be repeated for the theorem below and, hence, we omit the details.

Theorem 2.21. *Let f be Hölder continuous of order α; that is,*

$$|f(\theta+h) - f(\theta)| \le |h|^\alpha. \tag{2.59}$$

Then

$$|c_n(f)| \le \frac{C}{2^{1+\alpha}}|n|^{-\alpha}. \tag{2.60}$$

Theorem 2.22. *Let $f \in C^1(S^1)$. Then $c_n(f') = (2\pi in)c_n(f)$, and*

$$|c_n(f)| \le \frac{\|f'\|_1}{2\pi|n|}. \tag{2.61}$$

Proof. We have

$$c_n(f') = \int_0^1 f'(\theta)e^{-2\pi in\theta}\,d\theta = \left[f(\theta)e^{-2\pi in\theta}\right]_0^1 + 2\pi in\int_0^1 f(\theta)e^{-2\pi in\theta}\,d\theta = (2\pi in)c_n(f). \tag{2.62}$$

Hence, $|c_n(f)| = \frac{1}{2\pi|n|}|c_n(f')| \le \frac{\|f'\|_1}{2\pi|n|}$. ∎

Corollary 2.23. *Let $f \in C^k(S^1)$. Then $c_n(f^{(k)}) = (2\pi in)^k c_n(f)$, and*

$$|c_n(f)| \leq \frac{C}{(2\pi\,|n|)^k}. \tag{2.63}$$

Corollary 2.24. *Let $f \in C^\infty(S^1)$. Then $\lim_{|n|\to\infty} |n|^k\,|c_n(f)| = 0$ for all $k \geq 0$.*

2.4 Hilbert Space and Orthonormality

October 1st.

Definition 2.25. Let H be a vector space over $\mathbb{C}$. An *inner product* on H is a map $\langle\cdot,\cdot\rangle : H \times H \to \mathbb{C}$ such that for all $x, y, z \in H$ and $\lambda \in \mathbb{C}$, the following hold.

1. $\langle x, y\rangle = \overline{\langle y, x\rangle}$.
2. $\langle x + y, z\rangle = \langle x, z\rangle + \langle y, z\rangle$.
3. $\langle \lambda x, y\rangle = \lambda\langle x, y\rangle$.
4. $\langle x, x\rangle > 0$ if $x \neq 0$.

The norm induced by the inner product is defined as $\|x\| = \sqrt{\langle x, x\rangle}$ for $x \in H$. If H is complete with respect to the metric $d(x, y) = \|x - y\|$, then H is called a *Hilbert space*.

Example 2.26. Let $H = \mathbb{C}^n$, and define the inner product $\langle x, y\rangle = \sum_{j=1}^n x_j\overline{y_j}$ for $x, y \in \mathbb{C}^n$. Then we know that $(\mathbb{C}^n, \langle\cdot,\cdot\rangle)$ is a valid inner product space. We claim that this space is also complete. Let $(x^{(k)})_{k\geq 1}$ be a Cauchy sequence in $\mathbb{C}^n$. Then for each $j = 1, \ldots, n$, the sequence $(x_j^{(k)})_{k\geq 1}$ is a Cauchy sequence in $\mathbb{C}$, which is complete, so there exists $x_j \in \mathbb{C}$ such that $x_j^{(k)} \to x_j$ as $k \to \infty$. Set $x = (x_1, \ldots, x_n) \in \mathbb{C}^n$. Then $x^{(k)} \to x$ as $k \to \infty$, and hence $\mathbb{C}^n$ is complete.

Example 2.27. Let $H = L^2[0, 1]$, and define the inner product $\langle f, g\rangle = \int_0^1 f(x)\overline{g(x)}\,dx$ for $f, g \in L^2[0, 1]$. Then $(L^2[0, 1], \langle\cdot,\cdot\rangle)$ is a valid inner product space. We claim that this space is also complete. Let $(f_k)_{k\geq 1}$ be a Cauchy sequence in $L^2[0, 1]$. Then there exists a subsequence $(f_{k_j})_{j\geq 1}$ such that $\|f_{k_{j+1}} - f_{k_j}\|_2 < 2^{-j}$ for all $j \geq 1$. Define $g_j = f_{k_{j+1}} - f_{k_j}$ for $j \geq 1$. Then

$$\sum_{j=1}^{\infty} \|g_j\|_2 \leq \sum_{j=1}^{\infty} 2^{-j} = 1. \tag{2.64}$$

Hence, by the triangle inequality, we have

$$\left\|\sum_{j=1}^{N} g_j\right\|_2 \leq \sum_{j=1}^{N} \|g_j\|_2 \leq 1, \tag{2.65}$$

so that $\sum_{j=1}^N g_j$ is bounded in $L^2[0, 1]$. By the completeness of $L^2[0, 1]$, there exists $g \in L^2[0, 1]$ such that $\sum_{j=1}^N g_j \to g$ as $N \to \infty$. Then $f_{k_N} = f_{k_1} + \sum_{j=1}^{N-1} g_j \to f_{k_1} + g$ as $N \to \infty$, so that $(f_k)_{k\geq 1}$ converges to $f = f_{k_1} + g \in L^2[0, 1]$. Hence $L^2[0, 1]$ is complete.

Also recall that two vector $x, y \in H$ are said to be orthogonal if $\langle x, y\rangle = 0$. A set of vectors $\{x_j\}_{j\in J}$ is said to be orthonormal if $\langle x_j, x_k\rangle = \delta_{jk}$ for all $j, k \in J$. Any orthonormal set of vectors is linearly independent.

Given any inner product space, possibly infinite dimensional, take the collection of orthonormal sets. Note that $\{\frac{x}{\|x\|}\}$ is a non-empty orthonormal set for $x \neq 0$, so this collection is non-empty. Partially order this collection via inclusion. Then any chain, here, has a least upper bound, the union of all collections in the chain. Thus, by Zorn's lemma, there exists a maximal orthonormal set $\{e_j\}_{j\in J}$. This set is called an *orthonormal basis* of H. However, the set itself may not be a basis in the usual sense, since it may not span H.

For example, consider the space $H = \ell^2(\mathbb{N})$, the space of all complex-valued square-summable sequences, and let $\{\delta_m\}_{m\in\mathbb{N}}$ be the standard orthonormal basis of H, where $\delta_m = (0, \ldots, 0, 1, 0, \ldots)$ with 1 in the m-th position. Then $\{\delta_m\}_{m\in\mathbb{N}}$ is an orthonormal basis of H, but it does not span H, since there exist elements in H that cannot be expressed as finite linear combinations of the δ_m's. For instance, the element $(\frac{1}{n})_{n\in\mathbb{N}} \in H$ cannot be expressed as a finite linear combination of the δ_m's. What is true is that the span of $\{e_j\}_{j\in J}$ is dense in H, so that for any $x \in H$ and $\varepsilon > 0$, there exists a finite linear combination of the e_j's that approximates x within ε.

Theorem 2.28. *Let $\{e_0, e_1, \ldots\}$ be an orthonormal basis of a Hilbert space H, and let $s_n = \sum_{k=0}^{n}\langle x, e_k\rangle e_k$ and $t_n = \sum_{k=0}^{n} b_k e_k$ for $x \in H$ and $b_k \in \mathbb{C}$. Then*

$$\|x - s_n\| \leq \|x - t_n\|, \tag{2.66}$$

and equality holds if and only if $b_k = \langle x, e_k\rangle$ for all $k = 0, 1, \ldots, n$.

In other words, the theorem states that the orthogonal projection of x onto the span of $\{e_0, \ldots, e_n\}$ is the best approximation of x in that subspace, and the coefficients of this projection are given by the inner products $\langle x, e_k\rangle$.

Proof. We have

$$\|x - t_n\|^2 = \langle x - t_n, x - t_n\rangle = \langle x, x\rangle - \sum_{k=0}^{n} b_k\langle x, e_k\rangle - \sum_{k=0}^{n}\overline{b_k}\langle e_k, x\rangle + \sum_{j,k=0}^{n} b_j\overline{b_k}\langle e_j, e_k\rangle. \tag{2.67}$$

Since $\{e_0, \ldots, e_n\}$ is orthonormal, we have $\langle e_j, e_k\rangle = \delta_{jk}$, so that the last term simplifies to $\sum_{k=0}^{n}|b_k|^2$. Thus,

$$\|x - t_n\|^2 = \|x\|^2 - 2\Re\left(\sum_{k=0}^{n} b_k\langle x, e_k\rangle\right) + \sum_{k=0}^{n}|b_k|^2. \tag{2.68}$$

Now, consider the expression $\sum_{k=0}^{n}(b_k - \langle x, e_k\rangle)\overline{(b_k - \langle x, e_k\rangle)}$, which is non-negative. Expanding this, we get

$$0 \leq \sum_{k=0}^{n}(|b_k|^2 - 2\Re(b_k\overline{\langle x, e_k\rangle}) + |\langle x, e_k\rangle|^2) = \sum_{k=0}^{n}|b_k|^2 - 2\Re\left(\sum_{k=0}^{n} b_k\overline{\langle x, e_k\rangle}\right) + \sum_{k=0}^{n}|\langle x, e_k\rangle|^2. \tag{2.69}$$

Rearranging, we obtain

$$2\Re\left(\sum_{k=0}^{n} b_k\overline{\langle x, e_k\rangle}\right) \leq \sum_{k=0}^{n}|b_k|^2 + \sum_{k=0}^{n}|\langle x, e_k\rangle|^2. \tag{2.70}$$

But we also have

$$\|x - s_n\|^2 = \|x\|^2 - 2\Re\left(\sum_{k=0}^{n}\langle x, e_k\rangle\overline{\langle x, e_k\rangle}\right) + \sum_{k=0}^{n}|\langle x, e_k\rangle|^2 = \|x\|^2 - \sum_{k=0}^{n}|\langle x, e_k\rangle|^2. \tag{2.71}$$

Therefore,

$$\|x - s_n\|^2 \leq \|x - t_n\|^2, \tag{2.72}$$

which proves the theorem. ∎

Theorem 2.29. *Let $\{e_0, e_1, \ldots\}$ be an orthonormal basis of a Hilbert space H.*

1. *The series $\sum_{n=0}^{\infty} |\langle x, e_n\rangle|^2$ converges for all $x \in H$, and satisfies the* Bessel's inequality *$\sum_{n=0}^{\infty} |\langle x, e_n\rangle|^2 \leq \|x\|^2$.*
2. *The equation $\sum_{n=0}^{\infty} |\langle x, e_n\rangle|^2 = \|x\|^2$ holds for some $x \in H$ if and only if $\lim_{n\to\infty} \|x - s_n\| = 0$, where $s_n = \sum_{k=0}^{n} \langle x, e_k\rangle e_k$.*

Proof. Let $s_n = \sum_{k=0}^{n} \langle x, e_k\rangle e_k$, and let $t_n = 0$. Then by the previous theorem, we have

$$\|x - s_n\|^2 \leq \|x\|^2 . \tag{2.73}$$

Expanding $\|x - s_n\|^2$, we get

$$\|x - s_n\|^2 = \langle x - s_n, x - s_n\rangle = \|x\|^2 - 2\Re\left(\sum_{k=0}^{n} \langle x, e_k\rangle \overline{\langle x, e_k\rangle}\right) + \sum_{k=0}^{n} |\langle x, e_k\rangle|^2 = \|x\|^2 - \sum_{k=0}^{n} |\langle x, e_k\rangle|^2 . \tag{2.74}$$

Therefore,

$$0 \leq \|x - s_n\|^2 = \|x\|^2 - \sum_{k=0}^{n} |\langle x, e_k\rangle|^2 , \tag{2.75}$$

$$\sum_{k=0}^{n} |\langle x, e_k\rangle|^2 \leq \|x\|^2 . \tag{2.76}$$

Since this holds for all n, the series $\sum_{n=0}^{\infty} |\langle x, e_n\rangle|^2$ converges and satisfies Bessel's inequality.

For the second part, if $\lim_{n\to\infty} \|x - s_n\| = 0$, then taking the limit in the previous inequality gives

$$0 = \lim_{n\to\infty} \|x - s_n\|^2 = \|x\|^2 - \lim_{n\to\infty} \sum_{k=0}^{n} |\langle x, e_k\rangle|^2 , \tag{2.77}$$

which implies that $\sum_{n=0}^{\infty} |\langle x, e_n\rangle|^2 = \|x\|^2$. Conversely, if $\sum_{n=0}^{\infty} |\langle x, e_n\rangle|^2 = \|x\|^2$, then

$$\lim_{n\to\infty} \|x - s_n\|^2 = \|x\|^2 - \lim_{n\to\infty} \sum_{k=0}^{n} |\langle x, e_k\rangle|^2 = 0, \tag{2.78}$$

which means that $\lim_{n\to\infty} \|x - s_n\| = 0$. ∎

Note that if $\{e_0, e_1, \ldots\}$ is an orthonormal basis of H, then $\operatorname{span}\{e_0, e_1, \ldots\}$ is dense in H. Hence, for any $x \in H$, we an write it as $x = \sum_{n=0}^{\infty} \langle x, e_n\rangle e_n$, where the series converges in the norm of H, with $\|x\|^2 = \sum_{n=0}^{\infty} |\langle x, e_n\rangle|^2$. This is *Parseval's identity*, which is a generalization of the Pythagorean theorem to infinite-dimensional Hilbert spaces. For the Hilbert space $L^2[0,1]$, the set $\{e^{2\pi i n x}\}_{n\in\mathbb{Z}}$ is an orthonormal basis, and hence for any $f \in L^2[0,1]$, we can write

$$f(x) = \sum_{n\in\mathbb{Z}} c_n(f) e^{2\pi i n x}, \qquad c_n(f) = \int_0^1 f(x) e^{-2\pi i n x}\, dx, \tag{2.79}$$

October 5th. Thus, if H is a Hilbert space with a countable orthonormal basis $\{e_0, e_1, \ldots\}$, then H is isometrically isomorphic to $\ell^2(\mathbb{N})$ via the map

$$H \to \ell^2(\mathbb{N}), \qquad x \mapsto (\langle x, e_0\rangle, \langle x, e_1\rangle, \ldots). \tag{2.80}$$

Theorem 2.30 (The *Riesz-Fischer theorem*)**.** *Let $\{\varphi_0, \varphi_1, \ldots\}$ be an orthonormal system for $L^2[0,1]$, and let $\{c_n\}_{n=0}^{\infty}$ be a square-summable sequence of complex numbers. Then there exists a unique $f \in L^2[0,1]$ such that $c_n = \langle f, \varphi_n\rangle$ for all $n \geq 0$, and $\|f\|^2 = \sum_{n=0}^{\infty} |c_n|^2$.*

Proof. Define $s_n = \sum_{k=0}^{n} c_k \varphi_k$. Then $\|s_n - s_m\|^2 = \sum_{k=m+1}^{n} |c_k|^2$ for $n > m$, which tends to 0 as $m, n \to \infty$ since $\{c_n\}$ is a square-summable sequence. Hence, $(s_n)_{n\geq 0}$ is a Cauchy sequence in $L^2[0,1]$, which is complete, so there exists $f \in L^2[0,1]$ such that $s_n \to f$ as $n \to \infty$. Then

$$\langle f, \varphi_n \rangle = \lim_{m\to\infty} \langle s_m, \varphi_n \rangle = \lim_{m\to\infty} c_n = c_n, \tag{2.81}$$

and

$$\|f\|^2 = \lim_{n\to\infty} \|s_n\|^2 = \lim_{n\to\infty} \sum_{k=0}^{n} |c_k|^2 = \sum_{n=0}^{\infty} |c_n|^2 . \tag{2.82}$$

■

Theorem 2.31 (*Carleson's theorem*)**.** *Let $f \in L^2[0,1]$. Then the Fourier series of f converges to f pointwise almost everywhere on $[0,1]$.*

For a function $f \in L^1[0,1]$, we can define the n-th partial sum of the Fourier series of f as

$$(s_N f)(\theta) = \sum_{n=-N}^{N} \hat{f}(n) e^{2\pi i n\theta}. \tag{2.83}$$

If we define the Dirichlet kernel $D_N(\theta) = \sum_{n=-N}^{N} e^{2\pi i n\theta}$, then we can write

$$(s_N f)(\theta) = \sum_{n=-N}^{N} \int_0^1 f(t) e^{-2\pi i n(\theta - t)}\, dt = (f * D_N)(\theta). \tag{2.84}$$

Some properties of D_N include $\int_0^1 D_N(\theta)\, d\theta = 1$ and $\|D_N\|_1$ grows as $\frac{4}{\pi^2} \log N$. From here, also arises Fejér's kernel, defined as $\sigma_N f = \frac{1}{N} \sum_{n=0}^{N-1} s_n f$, which is the Cesàro mean of the partial sums of the Fourier series. In particular, denoting $F_N = \frac{1}{N} \sum_{n=0}^{N-1} D_k = \frac{1}{N} \left(\frac{\sin(N\pi t)}{\sin(\pi t)} \right)^2$, we have $\sigma_N f = f * F_N$. These F_n's are non-negative, integrate to 1, and for $\delta \leq |t| \leq \frac{1}{2}$, we have

$$F_N(t) \leq \frac{1}{N \sin^2(\pi\delta)} \to 0 \quad \text{as } N \to \infty; \tag{2.85}$$

the mass concentrates at 0 as $N \to \infty$. Also, rearranging the terms in F_N gives

$$F_{N+1}(t) = \sum_{n=-N}^{N} \left(1 - \frac{|n|}{N}\right) e^{2\pi i n t}. \tag{2.86}$$

2.4.1 Pointwise Convergence of Fourier Series

October 8th.

We will prove this in a series of steps. Let $f \in L^1(S^1)$ and $x_0 \in S^1$.

1. We have

$$(s_N f(x_0) - f(x_0)) = \int_{-1/2}^{1/2} (f(x_0 - t) - f(x_0)) D_N(t)\, dt. \tag{2.87}$$

   To show this, simply write $s_N f(x_0) = (f * D_N)(x_0) = \int_{-1/2}^{1/2} f(x_0 - t) D_N(t)\, dt$, and then use the fact that $\int_{-1/2}^{1/2} D_N(t)\, dt = 1$.

2. Dini's criterion: suppose

$$\int_{|t|<\delta} \frac{|f(x_0 - t) - f(x_0)|}{|t|}\, dt < \infty \tag{2.88}$$

for some $\delta > 0$. Then $s_N f(x_0) \to f(x_0)$ as $N \to \infty$. This can be done as

$$|s_N f(x_0) - f(x_0)| = \left| \int_{-1/2}^{1/2} \frac{f(x_0 - t) - f(x_0)}{\sin \pi t} \sin(2N+1)\pi t \, dt \right| \tag{2.89}$$

which goes to zero as $N \to \infty$, after bounding the integral over $|t| < \delta$ and $|t| \geq \delta$ separately.

3. If $|f(x) - f(x_0)| \leq C\,|x - x_0|^\alpha$ for some $\alpha > 0$, then $s_N f(x_0) \to f(x_0)$ as $N \to \infty$. This follows from Dini's criterion, since

$$\int_{|t|<\delta} \frac{|f(x_0 - t) - f(x_0)|}{|t|} \, dt \leq C \int_{|t|<\delta} |t|^{\alpha - 1} \, dt = \frac{C}{\alpha} \delta^\alpha < \infty. \tag{2.90}$$

4. If f has one-sided limits $f(x_0^\pm)$, and one-sided derivatives at x_0, then $s_N f(x_0) \to \frac{1}{2}(f(x_0^+) + f(x_0^-))$ as $N \to \infty$. Indeed,

$$\int_{|t|<\delta} \frac{\left|f(x_0 - t) - f(x_0^-)\right|}{|t|} \, dt < \infty, \qquad \int_{|t|<\delta} \frac{\left|f(x_0 + t) - f(x_0^+)\right|}{|t|} \, dt < \infty, \tag{2.91}$$

and the result follows.

5. If $f \equiv 0$ in a neighbourhood of x_0, then $s_N f(x_0) \to 0$ as $N \to \infty$. This follows from the previous step, since $f(x_0^\pm) = 0$ and the one-sided derivatives are also zero.

6. If f is Lipschitz continuous, then $s_N f(x_0) \to f(x_0)$ uniformly. To this end, note that if f is Lipschitz continuous, then there exists a constant $L > 0$ such that $|f(x) - f(y)| \leq L\,|x - y|$ for all $x, y \in S^1$. Hence, for any $x_0 \in S^1$ and $\delta > 0$, we have

$$\int_{|t|<\delta} \frac{|f(x_0 - t) - f(x_0)|}{|t|} \, dt \leq L \int_{|t|<\delta} dt = 2L\delta < \infty. \tag{2.92}$$

This shows that the integral is finite, and hence the convergence is uniform.

Chapter 3

SPACE OF CONTINUOUS FUNCTIONS

3.1 The Space $C[0,1]$

October 22nd.

Theorem 3.1. *The space $C[0,1]$ is Banach.*

Proof. Let $\{f_n\}_{n\geq 1}$ be a Cauchy sequence in $(C[0,1], \|\cdot\|_\infty)$. For each $x \in [0,1]$, the sequence $\{f_n(x)\}_{n\geq 1}$ is Cauchy in $\mathbb{C}$, since

$$|f_n(x) - f_m(x)| \leq \|f_n - f_m\|_\infty . \tag{3.1}$$

Since $\mathbb{C}$ is complete, there exists a function $f : [0,1] \to \mathbb{C}$ such that $f_n(x) \to f(x)$ for each $x \in [0,1]$.

We show that $f_n \to f$ uniformly on $[0,1]$. Let $\varepsilon > 0$. Since $\{f_n\}_{n\geq 1}$ is Cauchy, there exists $N \in \mathbb{N}$ such that

$$\|f_n - f_m\|_\infty < \varepsilon \tag{3.2}$$

for all $n, m \geq N$. Fix $n \geq N$ and let $m \to \infty$. Then, for every $x \in [0,1]$,

$$|f_n(x) - f(x)| \leq \varepsilon. \tag{3.3}$$

Taking the supremum over $x \in [0,1]$, we obtain $\|f_n - f\|_\infty \leq \varepsilon$. Hence, $f_n \to f$ uniformly on $[0,1]$. Since the uniform limit of continuous functions is continuous, we have $f \in C[0,1]$. Therefore, $\{f_n\}_{n\geq 1}$ converges to f in $(C[0,1], \|\cdot\|_\infty)$, and so $(C[0,1], \|\cdot\|_\infty)$ is a Banach space. ■

The space $C[0,1]$, however, is not compact. To see this, consider the sequence of functions $\{f_n\}_{n\geq 1}$ defined by $f_n(x) = n$ for $x \in [0,1]$. Then, $\|f_n\|_\infty = n \to \infty$ as $n \to \infty$, and so $\{f_n\}_{n\geq 1}$ has no convergent subsequence in $(C[0,1], \|\cdot\|_\infty)$. Now consider the unit ball $(C[0,1])_1 = \{f \in C[0,1] : \|f\|_\infty \leq 1\}$. The sequence $\{g_n\}_{n\geq 1}$ defined by $g_n(x) = x^n$ is contained in $(C[0,1])_1$, but it has no convergent subsequence in $(C[0,1], \|\cdot\|_\infty)$. Therefore, the unit ball $(C[0,1])_1$ is not compact in $(C[0,1], \|\cdot\|_\infty)$.

The *Heine-Borel theorem* states that $C(K)$, the space of continuous functions on a compact Hausdorff space K, is compact if and only if K is finite.

Definition 3.2. Let K be a compact metric space. A family of functions $F \subseteq C(K)$ is said to be

1. *uniformly bounded* if there exists $M > 0$ such that $\|f\|_\infty \leq M$ for all $f \in F$.
2. *equicontinuous* if for every $\varepsilon > 0$, there exists $\delta > 0$ such that $|f(x) - f(y)| < \varepsilon$ for all $f \in F$ and all $x, y \in K$ with $d(x,y) < \delta$.

Theorem 3.3 (The *Arzelà-Ascoli theorem*)**.** *Let K be a compact metric space. A family of functions $F \subseteq C(K)$ is relatively compact in $(C(K), \|\cdot\|_\infty)$ if and only if F is uniformly bounded and equicontinuous.*

Proof. First suppose that F is relatively compact. Since its closure is compact, we may replace F by its closure and assume that F is compact. If F is not uniformly bounded, then there exists a sequence $\{f_n\}_{n\geq 1} \subseteq F$ such that $\|f_n\|_\infty \to \infty$ as $n \to \infty$. By compactness, some subsequence $\{f_{n_k}\}_{k\geq 1}$ converges to a function $f \in F$. Hence,

$$\big|\|f_{n_k}\|_\infty - \|f\|_\infty\big| \leq \|f_{n_k} - f\|_\infty \to 0, \tag{3.4}$$

which is a contradiction. Therefore, F is uniformly bounded.

Suppose that F is not equicontinuous. Then there exist $\varepsilon > 0$, a sequence $\{f_n\}_{n\geq 1} \subseteq F$, and points $x_n, y_n \in K$ such that

$$d(x_n, y_n) < \frac{1}{n} \quad \text{and} \quad |f_n(x_n) - f_n(y_n)| \geq \varepsilon. \tag{3.5}$$

By compactness, after passing to a subsequence, $f_n \to f$ uniformly for some $f \in F$. Passing to a further subsequence, we may assume that $x_n \to x$ for some $x \in K$. Since $d(x_n, y_n) \to 0$, we also have $y_n \to x$. Thus,

$$|f_n(x_n) - f_n(y_n)| \leq |f_n(x_n) - f(x_n)| + |f(x_n) - f(y_n)| + |f(y_n) - f_n(y_n)| \to 0, \tag{3.6}$$

which is a contradiction. Hence, F is equicontinuous.

Conversely, suppose that F is uniformly bounded and equicontinuous. Let $\{f_n\}_{n\geq 1}$ be a sequence in F. Since K is a compact metric space, it has a countable dense subset $\{x_j\}_{j\geq 1}$. The sequence $\{f_n(x_1)\}_{n\geq 1}$ is bounded in $\mathbb{C}$, so it has a convergent subsequence. From this subsequence, choose a further subsequence for which $\{f_n(x_2)\}_{n\geq 1}$ converges, and continue this process. The diagonal sequence, which we denote by $\{g_n\}_{n\geq 1}$, has the property that $\{g_n(x_j)\}_{n\geq 1}$ converges for every $j \geq 1$.

Let $\varepsilon > 0$. By equicontinuity, there exists $\delta > 0$ such that

$$|f(x) - f(y)| < \frac{\varepsilon}{3} \tag{3.7}$$

for all $f \in F$ and all $x, y \in K$ with $d(x, y) < \delta$. By compactness, there exist $z_1, \ldots, z_m \in K$ such that

$$K \subseteq \bigcup_{i=1}^{m} B(z_i, \delta/2). \tag{3.8}$$

Since $\{x_j\}_{j\geq 1}$ is dense, choose $x_{j_i} \in B(z_i, \delta/2)$ for each i. Then, for every $x \in K$, there exists i such that $d(x, x_{j_i}) < \delta$. Since $\{g_n(x_{j_i})\}_{n\geq 1}$ converges for each i, there exists $N \in \mathbb{N}$ such that

$$|g_n(x_{j_i}) - g_m(x_{j_i})| < \frac{\varepsilon}{3} \tag{3.9}$$

for all $n, m \geq N$ and $1 \leq i \leq m$. Therefore, for all $n, m \geq N$ and $x \in K$,

$$|g_n(x) - g_m(x)| \leq |g_n(x) - g_n(x_{j_i})| + |g_n(x_{j_i}) - g_m(x_{j_i})| + |g_m(x_{j_i}) - g_m(x)| < \varepsilon. \tag{3.10}$$

Hence, $\{g_n\}_{n\geq 1}$ is uniformly Cauchy. By the completeness of $\mathbb{C}$, it converges uniformly to a function $g : K \to \mathbb{C}$. The uniform limit of continuous functions is continuous, so $g \in C(K)$. Thus every sequence in F has a subsequence converging in $C(K)$, and consequently F is relatively compact. ∎

October 26th.

Theorem 3.4 (*Peano's theorem*)**.** *Let $U \subseteq \mathbb{R}^{n+1}$ be an open subset. Let $f : U \to \mathbb{R}^n$ be a continuous function. Then the initial value problem*

$$\frac{dx}{dt} = f(t, x(t)), \quad x(t_0) = x_0 \tag{3.11}$$

has a solution $x(t)$ on (t_0-h, t_0+h) for some $h>0$.

Proof. Since U is open, there are $a, r > 0$ such that

$$[t_0-a, t_0+a] \times \overline{B(x_0, r)} \subseteq U. \tag{3.12}$$

Choose $h > 0$ so that $h < a$ and $h < r/(M+1)$, where

$$M = \sup\{\|f(t,z)\| : (t,z) \in [t_0-a, t_0+a] \times \overline{B(x_0,r)}\} < \infty. \tag{3.13}$$

For $n \geq 1$, put $t_k = t_0 + kh/n$ for $-n \leq k \leq n$ and define $x_n(t_0) = x_0$ and

$$x_n(t_{k+1}) = x_n(t_k) + \frac{h}{n} f(t_k, x_n(t_k)) \qquad (-n \leq k < n), \tag{3.14}$$

extending x_n linearly on each interval $[t_k, t_{k+1}]$. Inductively, every node lies in $\overline{B(x_0,r)}$, since

$$\|x_n(t_k) - x_0\| \leq M|t_k - t_0| \leq Mh < r.$$

Consequently, each x_n is well-defined and is M-Lipschitz on $[t_0-h, t_0+h]$. Thus the family $\{x_n\}$ is uniformly bounded and equicontinuous. By Arzelà–Ascoli, a subsequence, still denoted by $\{x_n\}$, converges uniformly to a function x.

For $s \in [t_k, t_{k+1})$, define $\tau_n(s) = t_k$. The polygonal construction gives

$$x_n(t) = x_0 + \int_{t_0}^{t} f\big(\tau_n(s), x_n(\tau_n(s))\big)\, ds. \tag{3.15}$$

Since $|\tau_n(s) - s| \leq h/n$, the uniform convergence $x_n \to x$ and the uniform continuity of f on the above compact set imply

$$f\big(\tau_n(s), x_n(\tau_n(s))\big) \longrightarrow f(s, x(s))$$

uniformly in s. Passing to the limit in the integral identity yields

$$x(t) = x_0 + \int_{t_0}^{t} f(s, x(s))\, ds.$$

The integrand is continuous, so the fundamental theorem of calculus gives $x'(t) = f(t, x(t))$ and $x(t_0) = x_0$. Hence x is a solution on (t_0-h, t_0+h). ∎

Index

algebra, 21
almost everywhere, 4
Arzelà-Ascoli theorem, 32

Bessel's inequality, 27

Carleson's theorem, 28
content zero, 4
convolution, 17

discrete Fourier basis, 1
discrete Fourier transform, 1
discrete Stone-Weierstrass theorem, 22
dominated convergence theorem, 10

equicontinuous, 31

Fatou's lemma, 11
Fourier coefficient, 11
Fourier series, 23
Fourier transform, 11, 15

Gamma function, 13

Heine-Borel theorem, 31
Hilbert space, 25

improperly Riemann integrable, 13
inner product, 25

Lebesgue integrable, 7
Lebesgue integrable function, 5
Lebesgue-Vitali theorem, 8

measurable function, 13
modulus of continuity, 24
monotone convergence theorem, 9

norm, 3

orthonormal basis, 26

Parseval's identity, 27
Peano's theorem, 32
Plancherel's theorem, 2
proto monotone convergence theorem, 4

Riemann-Lebesgue lemma, 23
Riesz-Fischer theorem, 27

self-adjoint, 22
step function, 3
Stone-Weierstrass theorem, 22

trigonometric polynomials, 23

uniformly bounded, 31
uniformly closed, 21

Weierstrass approximation theorem, 21